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Life cycle assessment of carbon capture and storage value chains: A systematic, critical literature review

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ABSTRACT

Quantification of greenhouse gas emissions is required for carbon capture and storage (CCS) projects to determine carbon offsets. Application of life cycle assessment (LCA) is the commonly applied framework for assessing life cycle impacts yet consensus and formal guidance on a rigorous approach is lacking for CCS. This study inter alia seeks to determine the level of consistency and uncertainty in reported LCAs. It follows a systematic literature-review protocol involving peer-reviewed articles on Scopus and Web-of-Science published between 2013 and 2024. A total of 33 articles were analysed according to a pre-defined set of criteria for data collection, analysis and reporting. We observed larger discrepancies in the general study characteristics, input values for energy demands and reported results. Low comparability across studies is caused by differences in methodological choices, functional unit and system boundaries. Less than half of the articles apply a full value chain perspective, and below one quarter includes projection of electricity carbon intensities. This review concludes that ambiguity arises in LCAs of CCS value chains due to repetitive omission of transport, storage and monitoring activities. We call for greater distinguishment between energy demands needed for capturing such as electricity use, heat generation, compression and liquefaction. Seeking standardization of carbon offsetting methodologies, we suggest including energy and material demands for transport, intermediate tank storage and injection processes along with drilling and post-injection monitoring as minimum system boundaries. Moreover, we propose the use of mechanistic modelling for validation of energy demands in life cycle inventories for prospective CCS projects.

Broader context

Carbon capture and storage (CCS) is broadly recognized as a sustainable alternative to reduce CO_2 emissions for achieving net-zero targets. Environmental impact assessment is performed to evaluate and justify the environmental advantages and carbon dioxide net-abatement potential. An example of methods fulfilling this goal is life cycle assessment (LCA) which provides a methodology for quantifying the life cycle environmental impacts across value chains and technologies. Carbon offsets are further issued for CO_2 emissions removed or offset as calculated by subtracting CCS project-related emissions from baseline emission values i.e., thereby determining the net-abatement potential of a CCS project across its entire value chain. Systematic and harmonized reporting across CCS projects and technologies are required to ensure consistency and documentation completeness in the claimed reduction potentials.

1. Introduction

As global temperatures continue to rise, significant and rapid efforts are needed to achieve large-scale emission reductions and keep within the Paris Agreement target of limiting global warming to 1.5 °C (IEA, 2024; Allen et al., 2022). Today, carbon capture and storage (CCS) of carbon dioxide (CO_2) is generally and globally perceived as a proven and effective way to isolate CO_2, either captured or removed from the atmosphere (IEA, 2024). By 2050, the International Energy Agency (IEA) estimates 5.9 Giga ton (Gt) of captured CO_2 to be stored annually compared to around 10 million ton (Mt) of CO_2 injected annually in 2022 (IEA, 2022). Transport and storage of CO_2 also play a vital role in the Net Zero Emissions by 2050 Scenario, underpinning a widespread deployment of carbon capture solutions, including direct air capture

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Nomenclature

Abbreviations

AC	activated carbon
AMP	2-amino-2-methyl-1-propanol
BECCS	bioenergy with carbon capture storage
CC	carbon credits
CCS	carbon capture and storage
CCU	carbon capture and utilization
CCUS	carbon capture utilization and storage
CDR	carbon dioxide removal
CO_2	carbon dioxide
DEA	diethanolamine
GHG	greenhouse gas
H_2	hydrogen
HCLC	hybrid coal chemical lopping
HPC	hot potassium carbonate
IGCC	internal gasification combined cycle
IL	ionic liquid-based [Bmim][Acetate]
LCA	life cycle assessment
MEE/MEEP	membrane-based capture proceses
MDEA	methyl diethanolamine
MEA	monoethanolamine
MSWI	municipal solid waste incineration
NGCC	natural gasification combined cycle
PZ	piperazine
SMR	steam methane reforming
SPPC	supercritical pulverized coal power plants
SubPC	subcritical pulverized coal power plants
VPSA	vacuum pressure swing adsorption

Units

CO_{2eq}	carbon dioxide equivalents
kWh	kilowatt hour
Mt.	million tonnes
MW	megawatt
MWh	megawatt hour
GWh	gigawatt hour
tCO_2	tonnes of carbon dioxide

(DAC) and bioenergy with CCS (IEA, 2024). Since 2023, the planned capacities for CO_2 transport and storage have significantly increased to an expected 260 Mt. of CO_2 annually (IEA, 2024). By 2030, pipeline infrastructure projects aim to reach around 615 Mt. of CO_2, exceeding previously planned capacities (IEA, 2024). In general, the positive outlook for the CCS industry is driven by the maturation and strengthening of market conditions globally through policy implementation and co-aligned deployment of CCS value chains by operators (IEA, 2024).

The level of climate change mitigation from CCS projects depends on the storage site capacity, the energy and material efficiency of industrial processes as well as leakages throughout the entire CCS value chain (Metz et al., 2005). In 2022, the Intergovernmental Panel for Climate Change (IPCC) reaffirmed CCS as a critical technology for mitigating climate change and achieving global climate goals (IPCC, 2022). Recent published studies of pioneering CCS value chains such as Varling et al. 2023, Emborg et al. 2024, and Burger et al. 2024 reports a positive CO_2 net-abatement potential, justifying its positive contribution to climate change mitigation (Varling et al., 2023a; Emborg et al., 2024; Burger et al., 2024a). Critical voices e.g., Masson-Delmotte et al. 2019, Fuss et al. 2018, and Rogelj et al. 2018 advocate CCS to be a gratuitous attempt to solve climate change issues, since the amount of CO_2 stored remains inadequate compared to actual, global storage requirements (Masson-Delmotte et al., 2019; Fuss et al., 2018; Rogelj et al., 2018). For instance, according to IPCC 2022, Creutzig et al. 2016, and Minx et al. 2018 bioenergy production with carbon capture and storage (BECCS) is done at the expense of increasing other environmental impacts, in particular changes in land use and freshwater eutrophication (IPCC, 2022; Creutzig et al., 2016; Minx et al., 2018).

In this study, we define CCS value chains as a set of coherent processes enabling the separation of CO_2 from industrial and energy-related sources, transportation and storage, resulting in the permanent isolation of CO_2 from the atmosphere. The capture of CO_2 may be applied to small and large point sources. These point sources often include major CO_2-emitting industries such as oil and gas processing, cement production, and power plants (Metz et al., 2005). Other activities may include CO_2-seperation from biomass energy facilities. Once captured, the CO_2 is compressed to either a gaseous or liquid phase before it is transferred or transported for interim and long-term storage. Today, different storage methods are available both onshore and offshore; hereunder, geological storage in subsurface formation, e.g., depleted oil and gas fields or saline aquifers. The potential technical methods for CO_2 transportation span from pipeline options (both onshore and offshore) to trucks and vessels of different capacity sizes. Planned and deployed CCS projects typically represent different structures of CCS value chains, including a varying degree of technological, temporal and geographical representativeness, as well as different process characteristics (IEA, 2022). Such variances are caused by several external factors such as existing infrastructure options, regulatory aspects, costs and overall removal potential, distances, public perception and technical capacity (Metz et al., 2005).

The study's goal is to provide a systematic, critical review of existing peer-reviewed articles reporting on the life cycle impacts of CCS value chains. This is seen in the light of the emergence of life cycle assessment (LCA) as a preferred methodology for quantifying the carbon dioxide removal (CDR) or carbon offsetting potential of DAC/BECCS (i.e., removing CO_2 from the atmosphere) and CCS (i.e., preventing CO_2 from being released directly into the atmosphere) projects. The aim of the literature review is to establish an overview of reported values provided in past and present literature and to identify gaps and potential shortcomings that may lead to inaccuracy in calculations of the removal and offsetting potential. As a benchmark, we rely peripherally on the minimum requirements and calculation recommendations set out in a number of introductory standards for the quantification of carbon removals, hereunder, the PURO Earth standard methodology for geological storage from bioenergy processes (PURO Earth, 2024), UNIDO's foundation for CCS in product life cycle assessment (UNIDO, 2024), and the European Commission's draft methodology for certification of industrial carbon removals (European Commission and Malins, 2025).

Our literature review provides an overview of the general study and LCA characteristics of the selected studies (Section 4.1). We critically analyze and compare the reported LCA inputs (Section 4.2) and outputs (Section 4.3) both qualitatively (according to key LCA aspects such as the functional unit, level of process inclusion, and inventory completeness) and quantitatively (comparison of input and output values across the selected studies and against theoretical values). Lastly, recommendations towards streamlining LCAs of CCS as well as strengthening in respect to the recently introduced standard methodologies for carbon removals are provided along with a discussion of the study limitations and recommended future work (Section 5).

2. Literature review

This Section briefly summarizes state-of-the-art in LCA of BE/CCS value chains as well as key findings from previously published scientific review articles discussing the application of LCA in CCS.

As of today, a wide range of literature reviews of studies assessing the environmental impact of CCS technologies have been conducted e.g.,

Garcia-Garcia et al. 2021, Müller et al. 2020, and Terlouw et al. 2021 (Cuéllar-Franca and Azapagic, 2015; Storrs et al., 2023; Chelvam and Hanafiah, 2023; Garcia-Garcia et al., 2021; Müller et al., 2020; Roberts and Stalker, 2020; Terlouw et al., 2021). It is our perception that most of these studies address the environmental performance from a techno-economic perspective or through an environmental due diligence demonstration (Storrs et al., 2023). Yet to some extent, techno-economic assessments (TEAs) are proven insufficient at identifying environmental challenges to CCS deployment mainly due to a lack of modelling beyond global warming impacts (Viebahn and Chappin, 2018). Alternatively, and as suggested by Terlouw et al. (2021) life-cycle assessment (LCA) methodologies present a more flexible framework, suitable for the modelling of environmental impacts of CDR technologies such as eutrophication, land use and water depletion. According to Chelvam and Hanafiah (2023), the tendency to apply LCA for CCS technologies is backed up by the increased number of articles published over the past decade which spiked in 2021 (Chelvam and Hanafiah, 2023). However, concerns have been raised within the scientific community, highlighting key challenges related to the LCA methodological and modelling choices. In a critical review of LCA of CDRs, Terlouw et al. (2021) finds that LCAs must be conducted with multiple environmental impact categories, while calling for more transparency regarding modelling choices and disclosure of life-cycle inventories. According to Wang et al. (2022), published LCA results demonstrate the presence of sharp trade-offs between climate change reductions and energy demand as well as water consumption for carbon capture technologies. Moreover, the need for harmonizing LCA methodologies was confirmed by Garcia-Garcia et al. (2021). The authors conducted a literature review on LCA applied to carbon capture and utilization (CCU) technologies, concluding that different LCA studies of similar CCU processes often resulted in divergent results. Moreover, LCA studies on CCU technologies were concluded as often incomparable due to e.g. variation in methodological choices, difference in the functional unit, i.e., the relative basis for which environmental impacts are assessed, or system boundaries (Müller et al., 2020). As a result, both the academic literature (Cuéllar-Franca and Azapagic, 2015; Naims, 2016) and the Scientific Advice Mechanism (SAM) of the European Commission (2018) called for specific guidelines for LCA on CCU technologies. We attempt to fill the current research gap and test if similar conclusions should be made for LCA application of CCS.

The complexity of CCS value chain characteristics underpins several requirements towards the methodological approach applied for quantification of BE/CCS projects' CO_2 removal or offsetting potential. First, life cycle impact assessment methodologies need to evaluate the performance of the entire value chains. Secondly, cradle-to-grave assessment of value chain activities needs to be quantified to justify the net reduction of CO_2 emissions across the entire lifespan of the CCS project. As CCS projects around the world progressively mature, standard methodologies for quantifying the CO_2 removal or offsetting potential are introduced. Most recently in March 2025, the European Commission published a draft technical specification for the certification of permanent carbon removals through DACCS and BECCS (referred to as the EU Carbon Removal Certification Framework (CRCF) methodologies (Office of the European Union L- P, Luxembourg L, 2024)), that aims to support the development of methodologies for assessing GHG emissions of permanent storage. In contrast to existing literature reviews, we contribute with an evaluation of how LCA has commonly been applied in the context of BE/CCS and compare this with well-known standard methods to pinpoint potential pitfalls or shortcomings to be aware of.

Historically, studies have tended to focus on the comparison of methods e.g., TEA vs LCA, or application within a specific industry or technology e.g., CCU vs CCS, or MEA vs. membrane-based capture. For instance, Storrs et al. focus on identifying critical challenges in CCUS projects with application of TEAs, Terlouw et al. (2021) focus on LCA application for a range of CDR technologies rather than offsetting technologies, whereas Robert et al. focus on monitoring technologies. Though the overall study scope of Cuéllar-Franca and Azapagic (2015) is similar, the authors focus on CCUS and the comparison between CCS and CCU, listing recommendations towards impact reduction and the comparison of impacts between the two technologies. Likewise, Mendoza et al. (2022a), and Müller et al. (2020) focus explicitly on CCU value chains or technologies, which in contrast leaves CCS relatively understudied albeit this technology is rapidly deployed. Given the rapid deployment of storage projects within recent years and considering the exponential increase in publications that centers on LCA of BE/CCS case studies, particularly from 2020 and onwards, we present an updated literature review which includes the most recent scientific articles. At the same time, a more stringent focus is placed on the dispersion within system boundaries and the inclusion and characterization of CCS processes than previously covered in literature.

3. Methods

Section 3.1 describes the protocol used for conducting literature research and analysis. Section 3.2 describes the study selection criteria and Section 3.3. the systematic selection of relevant publications. Section 3.4 describes the approach for validating the uncertainty of reported energy demands via mechanistic modelling.

3.1. Literature research protocol

A systematic literature review was conducted following the guideline outlined by Flávio et al. (2017), Glarou et al. (2020) and Pieroni et al. (2019), with the objective to identify and analyze articles and reports on life cycle environmental impact of CCS. The review methodology followed a systematic literature review protocol, comprising three main steps. See Fig. 1. First, the data collection (research question, source selection and study selection). Second, the data analysis (information extraction and result summarization), and third, the data reporting (dissemination of results). Two main research questions (RQ) with additional sub-questions were established to guide data collection and analysis. The full review protocol is shown in **SM.1**.

RQ1. How is LCA applied as a method for assessing the environmental impact associated with CCS processes and technologies?

i. To what extent are LCA consistently applied across CCS studies and how may this affect evaluation and comparability?

RQ2. What are the reported environmental impacts associated with CCS value chains?

i. To what extent are results consistently reported across CCS studies?
ii. To what extent does methodological inconsistency impact the reported results?

3.2. Review criteria and study selection

In respect to the research questions, specific search keywords were chosen according to four different elements (method, system, technology, and infrastructure-related keywords). The keywords were then assembled into one search string and applied to the Scopus and Web of Science (WoS) databases. A search filter was added to restrain on the publishing year (2013-2024) which is deemed a suitable timeframe considering the increase in planning and deployment of CCS projects within the past decade, language (English) and document type (e.g. open-source articles; see Table 1). The string was altered through three iterations and was considered ideal when returning sufficient hits (approximately 1000 hits) while also including five pre-selected key papers by Mendoza et al. (2022b), Müller et al. (2020), Quéheille et al. (2021), Terlouw et al. (2021), and Yuan et al. (2023). These publications

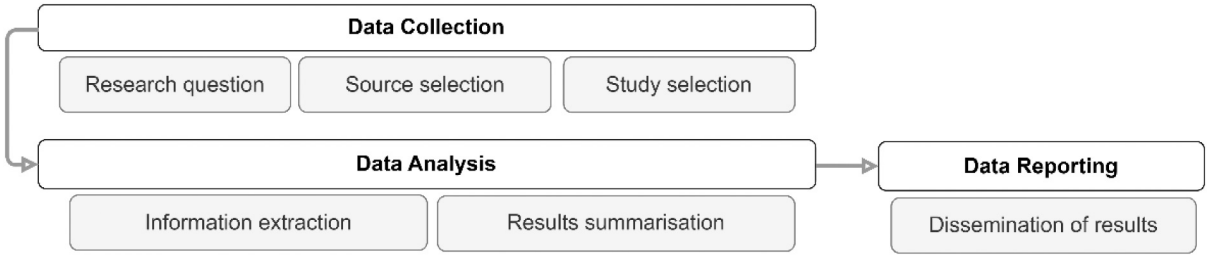


Fig. 1. Steps of the systematic literature review protocol applied for this study as adapted from Pieroni et al. (2019).

Table 1
Search parameters for the systematic literature review.

Database	Scopus, Web of Science
Fields	(1) title, keywords, abstract
Keywords (related terms)	*Method:* footprint*, impact*, quantification*, environment*, assess*, carbon, footprint, life cycle assessment*, LCA, value chain* *System:* offshore, subsea, depleted, platform, reservoir, carbon emitter* *Technology:* carbon*, carbon storage, carbon injection, monoethanolamine, MEA, conditioning, compression, carbon capture and storage*, CCS, monitoring *Infrastructure:* pipeline, vessel, ship, supply, tanker
Text language	English
Published Year	$2013 \leq Y \leq 2024$
Type of retrieved publications	Journal articles, conference papers, book chapters

were pre-selected based on their fulfillment of the two research questions and the number of citations. Two inclusion criteria (IC) were defined to guide the study selection process: (IC1) the studies must present the method or tool applied for the life cycle impact assessment of CCS technologies or operations, (IC2) the studies must be explicitly related to CCS. After running the search string, the publications were qualitatively selected according to three filters: filter 1 – reading title, abstract and keywords, filter 2 – reading introduction and conclusion, and filter 3 – complete reading of publication.

The literature search resulted in 2650 articles from two databases (see Table 2): Web of Science (1276 articles retrieved in July 2024) and Scopus (1905 articles retrieved in May 2023). After the abstract reading, 403 unique articles were included for further reading (filter 2) based on IC1. After reading the introduction and conclusion reading, 181 publications were included for full article reading (filter 3) based on IC2. At the final text analysis level, 33 publications were included for full text analysis if RQ 1 and 2 were answered. This selection was based on whether the articles answered either or both main research questions.

3.3. Evaluation of relevant literature

The goal of any LCA is to quantify and evaluate the environmental impacts of a product, system or service across its entire lifespan (Hauschild, 2018). The framework for LCA is based on international standards (International Organization for Standardization, 2006a; International Organization for Standardization, 2006b) with further guidance provided by ILCD (Wolf et al., 2010). The LCA characteristics applied in the reviewed literature are presented with respect to the methodological approach and the impact assessment. For the impact assessment, we focused on the European Commission's sixteen recommended impact categories (Directorate-General for Environment, 2021) including climate change, ozone depletion, eutrophication, acidification, human toxicity, respiratory inorganics, ionizing radiation, ecotoxicity, photochemical ozone formation, land use and resource depletion (Bassi et al., 2023; Directorate-General for Environment, 2021).

Table 2
Number of articles found and remaining after the study selection steps.

Steps	Number of articles	Study selection criteria
Filter 1 – Title, abstract and keywords	2650	–
Filter 2 – Introduction and conclusion	403	Duplicate removal + IC1
Filter 3 – Complete article reading	181	IC2
Final inclusion - Full text analysis	33	RQ1 and RQ2

The 33 articles selected for final inclusion were analysed according to the pre-defined list of criteria shown below. The complete list of answers to each criterion from the data analysis is available in **SM.1.**

- Description of the general study characteristics (e.g., scope of study, applied value chain perspective, number of CCS processes included and the level of process granularity)
- Application within industry (e.g., power plants, waste-to-energy plants, hydrogen production)
- LCA characteristics (e.g., documentation of modelling choices, number of scenarios, article and study type, presence of sensitivity and uncertainty analysis)
- Scenario projections and characteristics (e.g., time horizons, modelling techniques, documentation of methodology for scenario modelling)
- Quantitative measures (i.e., reported numerical input values for the LCA, generated output values and environmental impact categories coverage)
- Qualitative measures (i.e., reported input values as per value chain process)

In general, CCS routes from capturing CO_2 to permanent storage are highly project specific. The usual characteristic of a CO_2 management value chain often depicts the capturing of CO_2, transportation of CO_2 and either utilization or permanent storage of CO_2 in geological formations (IEA Handbook, 2023). See Fig. 2. We used the same requirements as mentioned by Terlouw et al. (2021) to determine when BE/CCS contributes to the removal or avoidance of CO_2. First, CCS projects must lead to permanent isolation of CO_2. Second, the climate change impact reported for the value chain of a CCS project must include all upstream and downstream GHG emissions. Third, the net reduction of a BE/CCS project must be lower than the total GHG emissions emitted throughout the CCS value chain.

Activities in the CCS value chains typically comprise four major steps: carbon capture (either from biogenic or post-combustion processes e.g., biogas production or cement production), transportation of compressed CO_2, permanent storage of CO_2 in geological formations either onshore or offshore, and storage monitoring (IEA, 2022; Global CCS Institute, 2024; Project Greensand, 2024). To evaluate the general study and LCA characteristics across the relevant publications, a generic boundary definition is set for post-combustion CCS and BECCS value chains based on the representation by Ansaloni et al. (2020), Varling

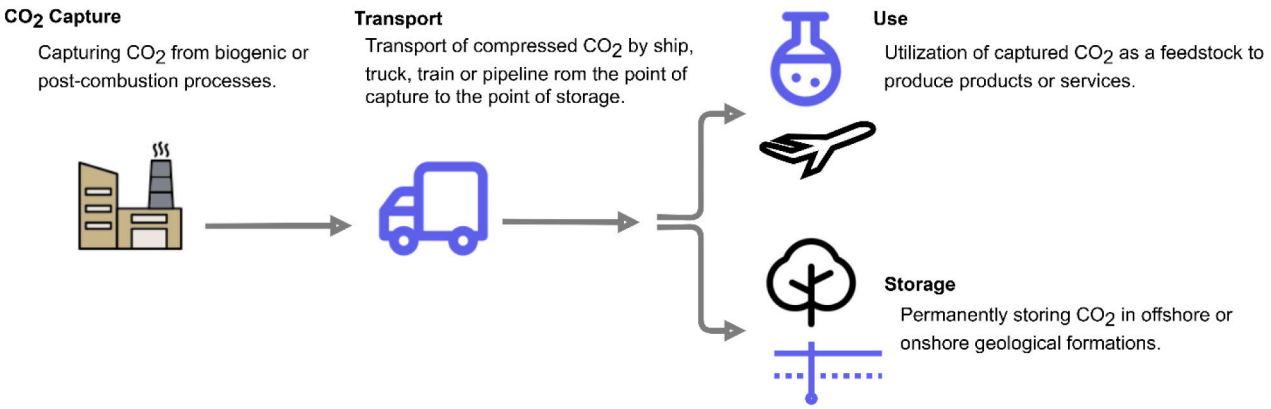


Fig. 2. Illustration of a potential CO_2 management value chain adopted from the International Energy Agency Carbon Capture Utilization and Storage Handbook (Metz et al., 2005).

et al. (2023b) and Emborg et al. (2024). See Fig. 3. Notably, CO_2 upgrade within BECCS may entail that CO_2 is already captured as part of the existing methanization process and thus capturing is excluded from the system boundaries e.g., in attributional modelling. Moreover, the backward pointing arrow indicates that there may be multiple transport and temporary storage links in a CCS value chain. Likewise, transport may entail both pipelines, onshore and/or offshore alternatives. Factors such as infrastructural limitations, traffic restrictions, geographical location and span as well as the amount and frequency of CO_2 to be shipped typically impact how the individual chain is designed.

3.4. Mechanistic modelling

We use mechanistic analysis to model the relationship between variables within a CCS project based on the different choices that can be made in technologies and designs. Mechanistic analysis is a theory-driven approach where the model is based on known or hypothesized mechanisms, e.g., physical or chemical processes. Rather than fitting data, the model seeks to explain the causality using mathematical models derived from domain knowledge. Often, mechanistic models include equations that describe real-world processes rather than just relying on statistical relationships. One example could be to use the laws of thermodynamics to model how fuel combustion leads to CO_2 emissions. In the following, we present some basic calculations and compare them with theoretical and state-of-the-art energy demands for CCS processes. Calculation procedures are provided in SM2. Results are presented in Section 4.3.

3.4.1. Theoretical energy demand for the capture process

The practical exergy demand of the process of separating CO_2 from a MEA solution can be obtained by dividing the minimum separation work by the efficiency of the separation process. The minimum work required

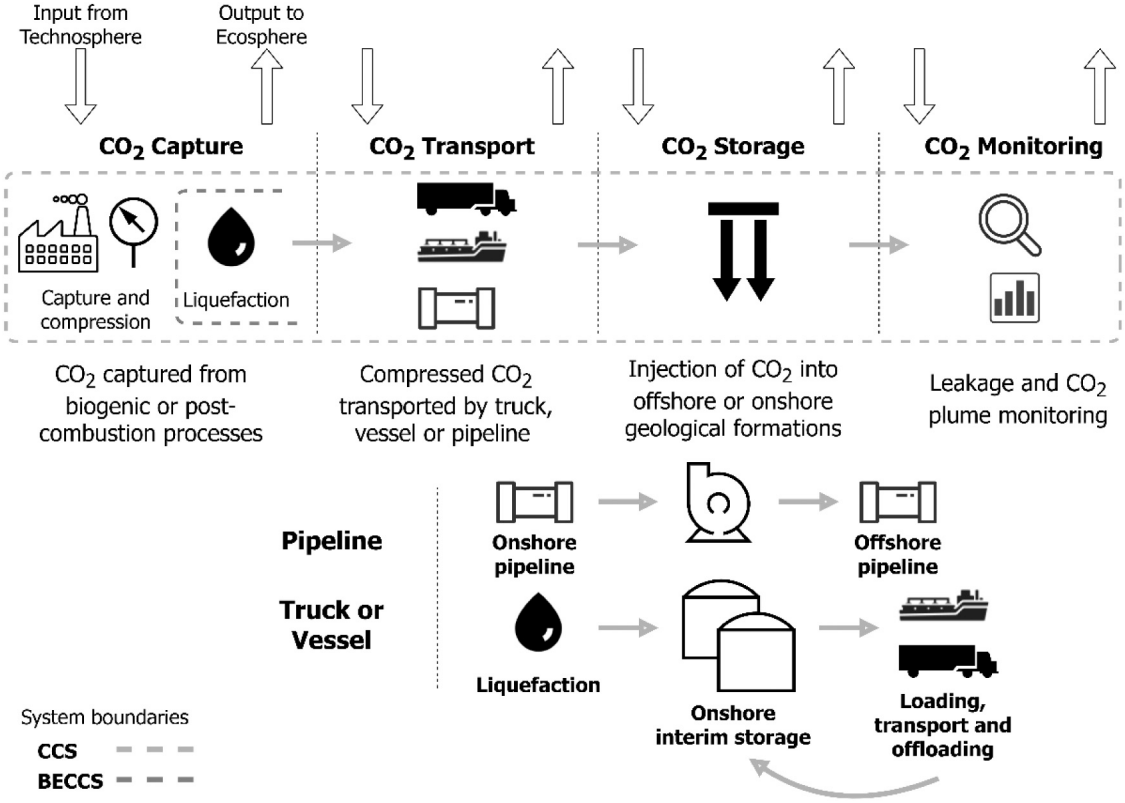


Fig. 3. Representation of the generalized (bioenergy) carbon capture and storage system boundaries and process definition used in the full text analysis of relevant publications.

to separate 95 % of CO_2 from a flue gas stream with a mole fraction of 10 vol% is calculated as 183 MJ/tCO_2. An alternative method for estimating the heat demand of the chemisorption process is the use of the amine to CO_2 ratio and calculating the flow rate of amine solution required to capture one ton of CO_2. For MEA solutions, a molar ratio (α) of 2.0 to 2.5 for amine to capture CO_2 is recommended. For an MEA to CO_2 molar ratio $\alpha = 2.2$ and a capture rate of 95 %, the reboiler heat duty can be estimated as 3080 MJ/tCO_2 (or 855 kwh/tCO_2), of which 231 MJ/tCO_2 is the heat loss by the hot gas streams. In practice, a higher value of α and thus a higher amount of heat is required to achieve a 95 % capture rate (Feron et al., 2020). The electricity demand of the blower for the transport of CO_2 into the absorption column at 1.2 bar can be estimated as 43 kWh/tCO_2 assuming a blower efficiency of 70 %. The pumping electricity consumption for the circulation of the liquid solvent is <3 kWh/tCO_2 for an assumed pressure increase of 5 bar.

3.4.2. Theoretical energy demand for liquefaction

For capture rates higher than 90 %, it is not possible to separate the CO_2 from a typical flue gas (i.e. 10-15 vol% CO_2) in a single stage. A capture rate of 80 % can be achieved by cooling a flue gas with 10 vol% CO_2 content to 155 K at 3 bars. This process requires 4.5 MJ electricity per ton of captured CO_2 (or 16 kWh/tCO_2) in using a Stirling cycle. The same concept could be applied for the liquefaction of a captured CO_2 stream to liquid phase for transport. The CO_2 stream is first pressurized to a moderate pressure above the triple point pressure of CO_2 at 5.18 bar (sometimes higher up to 70 bar), e.g. 6 to 17 bar and cooled down to the bubble point temperature. The required temperature is between -5.0 to 2.7 °C and therefore does not need a low temperature cooling system, resulting in higher COP usually higher than 1.0. For a COP of 1.2, we estimated the required electricity for the liquefaction of CO_2 (including both compression and refrigeration) to be 519 to 562 MJ/tCO_2 (or 144 to 156 kWh/tCO_2).

3.4.3. Theoretical energy demand for CO_2 compression

Compressing a CO_2 stream from atmospheric pressure (approximately 1 atm) to a supercritical pressure, typically 120 to 150 bar required for pipeline transportation, is a thermodynamically complex process requiring significant electrical energy input. The calculation of this energy consumption necessitates accounting for the real gas behavior of CO_2, which deviates significantly from ideal gas assumptions, particularly near its critical point (31.1 °C and 73.8 bar) and in the supercritical region. Eventually, the electrical energy consumption is calculated by dividing the total mechanical work by the efficiency of the compressor's driver, which is assumed to be an electric motor. For the intercoolers, we assume a temperature of 40 °C. We also assume that the compressor is used to compress the gas to 80 bar, and then a pump is used to increase the pressure further to 150 bar. For a compressor efficiency of 80 % and an electric driver efficiency of 90 %, the compression of CO_2 from 1 to 80 bar requires 387 MJ/tCO_2 (or 108 kWh/tCO_2) electricity. A multistage pump can be utilized to increase the pressure from 80 bar to 130 bar at 40 °C for supercritical transport. This operation (with a mechanical efficiency of 70 % and an electrical driver efficiency of 90 %) requires 14 MJ/tCO_2 (3.9 kWh/tCO_2) electricity.

3.4.4. Theoretical energy demand for pipeline transport

The pipeline diameter and thickness for the transport of a stream of supercritical CO_2 can be calculated by assigning pressure drop per unit length and the maximum pipe pressure. We assume 0.1 bar/km pressure drop and 130 bar for the maximum pressure. The mass of carbon steel for such a pipeline is between 66 and 246 ton/km. Based on the selected pressure drop per unit length, five pressure boosters (pumps) are required, which consume a total electricity amount of 70 MJ/ton CO_2 (19.4 kWh/tCO_2 per km) excluding 387 MJ/ton CO_2 (107 kWh/tCO_2) for the initial compression.

3.4.5. Theoretical energy demand for truck transport

The energy required for the transportation of a truck can be estimated by calculating the two major forces acting against its movement, viz. rolling resistance and aerodynamic drag. For a large truck that weighs 15 tons carrying 40 m^3 of liquid CO_2 at 15 bar (42.8 ton of CO_2), a 40 % efficient truck consumes 356 MJ/ton CO_2 for transporting over 1000 km (99 kWh/tCO2). This is equivalent to an energy consumption of 15.2 MJ/km or 37.5 L diesel/100 km, that is on par with the average fuel consumption of a fully loaded large truck. Conclusively, pipeline transportation consumes more energy but does not require CO_2 liquefaction.

3.4.6. Theoretical energy demand for reservoir injection

The energy requirement for the injection of supercritical CO_2 into a reservoir can be estimated by calculating the required pressure for the injection of CO_2 into a storage reservoir. For a 2000 m deep reservoir with a permeability of 1×10^{-14} m^2, an average CO_2 relative permeability of 0.2, a reservoir pressure of 200 bar at 1000 m from the injection well with a radius of 0.1 m and a 50 m perforated length, the injection pressure is calculated as 121.5 bar for the injection of 1 Mta CO_2. If CO_2 is transported by pipeline in supercritical condition and arrives at, e.g. 80 bars to the injection platform, the injection pump consumes 12.2 MJ electricity per ton CO_2 injected (3.4 kWh/tCO_2). If the electricity is provided on the platform by a gas or diesel turbine with 30 % efficiency, the fuel consumption is 40.8 MJ/ton CO_2 injected (11.3 kWh/tCO_2).

4. Results

This Section presents the main outcome of the literature review. Section 4.1 summarizes the general study and LCA characteristics of each study (Table 3) and the coverage of impact categories (Table 4). Section 4.2 summarizes the input values reported within the life cycle inventory (Table 5). Prevalence and the degree of harmonization across these scientific publications will be discussed further in Section 5. Section 4.3 presents the reported LCIA outputs for the climate change impact category where ambiguity is observed (Table 6). Section 4.4 compares the reported input values with theoretical values to check the uncertainty level of LCA inventories.

4.1. General study characteristics

Fig. 4 shows the development in peer-reviewed scientific articles

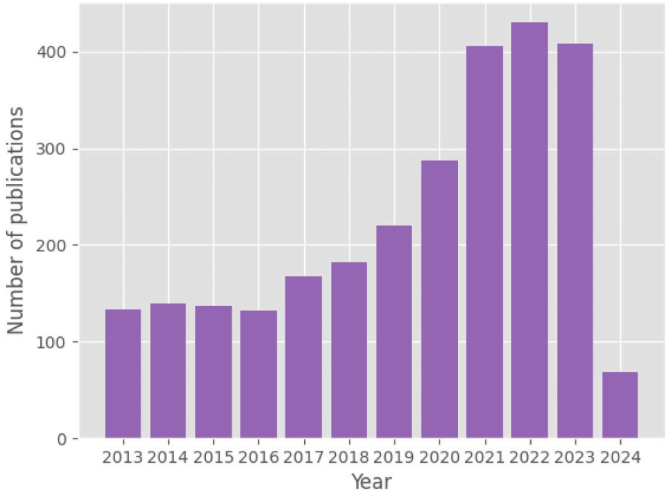


Fig. 4. Number of reported studies on carbon capture and storage value chains from 2013 until May 2024 across Scopus and Web of Science. The year 2024 includes publications for the first 5 months only.

Table 3
General study and life cycle assessment characteristics of the 33 reviewed studies.

Publication	Year	General study characteristics							LCA characteristics					
		CO_2 source	Capacity, Mt./year	Capture rate,	Capture scheme	System boundary	Unit processes	Temporal scope	Reference Flow	Energy scenarios	Sensitivity	Background modelling	Impact method	Midpoint categories
Volkart et al. (2013a)	2013	Post-combustion at IGCC/ NGCC power plants and cement production	N	90-95 %	MEA	Cradle-to-grave	C;T;S	2025-2050	1 kWh / 1 kg cement	Yes	Yes	ecoinvent 2.2	IPCC 2007 / ReCiPe v.1.07	7
Iribarren et al. (2013)	2013	Post-combustion at coal fired power plants	N	N	Various	Cradle-to-grave	C;T;S	N	1 kWh of net electricity	Yes	No	ecoinvent 2.0	CML 2001	6
Yujia et al. (2014)	2014	Post-combustion at coal fired power plant	N	60 %	MEA	Cradle-to-gate	C;T;S	2014	per kWh produced/ per ton CO_2	No	No	N	N	1
Lacy et al. (2015)	2015	Post-combustion NGCC power plant	1	N	MEA	Cradle-to-grave	C;T;S	2020-2050	1 kWh produced	Yes	Yes	ecoinvent 2.0	CML 2001	1
Troy et al. (2016)	2016	Post-combustion at ICGG power plant	N	90 %	MEA	Cradle-to-grave	C;T;S	N	1 kWh electricity produced	No	No	ecoinvent 2.2	ReCiPe 1.08	10
Štefanica et al. (2016)	2016	Lignite fired power plant	1.2	N	Ammonia scrubber	Cradle-to-gate	C	N	Production of 1 kWh	No	No	N	CML 2001	1
Navajas et al. (2019)	2017	Post-combustion from power generation	N	N	MEA/CLC	N	C;T;S	N	Production of 1 MWh	No	No	GaBi Pro. 8.7	EII	14
Young et al. (2019)	2019	Post-combustion at various processes	20.5-550	N	MEA	Cradle-to-gate	C;T;S	2014	1 kg CO_{2eq} abated	No	Yes	N	TRACI	6
Thorne et al. (2019)	2019	Theoretical HCLC plant	N	90-99 %	CLC	N	C	N	1 kWh of net electricity delivered to grid	No	Yes	ecoinvent 3.5	ReCiPe (H) 2016	4
An et al. (2019)	2019	Post-combustion from cement production	N	N	MEA	N	C;T;S	N	1 ton of cement	No	Yes	ecoinvent 3.0	CML 2001	10
Kim et al. (2019)	2019	Post-combustion at coal-fired power plant	0.7	95 %	Ammonia	Cradle-to-grave	C;T;S	N	0.7 Mt. CO_2 captured per year	No	Yes	ecoinvent 2.0	ReCiPe 2016	1
Yang et al. (2019)	2019	Post-combustion at pulverized coal power plant	N	90 %	MEA	Cradle-to-grave	C;T;S	N	1 MWh of net power produced	No	Yes	CLCD[b]	CML 2001	10
Zakrisson et al. (2023)	2020	Post/oxy combustion at coal fired power plant	N	75 %	AC absorption	Cradle-to-gate	C	N	Electricity production by the 250 MW coal power unit	No	No	N	ReCiPe 1.08 & Pareto analysis	18
Zakrisson et al. (2023)	2020	Pre-combustion at coal fired power plant	N	75-95 %	CaO looping & AC absorption	Cradle-to-grave	C	N	1 kWh of produced electricity	No	No	N	ReCiPe 1.08 & Pareto analysis	18
Bello et al. (2020)	2020	BECCS of bioethanol production.	N	N	MEA	Cradle-to-grave	C;T;S	N	1 km of distance travelled	No	Yes	Aspen Plus, ChemCAD	ReCiPe 1.1 Hierarchist	18
Antonini et al. (2020)	2020	Pre-combustion at H_2 production	N	90-98 %	VPSA & MDEA	Cradle-to-gate	C;T;S	N	Production of 1 MJ of compressed gaseous hydrogen	No	Yes	ecoinvent 3.5	IPCC 2013	8
Bisinella et al. (2021)	2021	Post-combustion at MSWI plant	0.3-1	85-95 %	MEA	Cradle-to-grave; no construction	C;T;S	2020-2030	1 ton of wet MSW	2030	Yes	ecoinvent 3.6	ReCiPe 2016	14

(*continued on next page*)

Table 3 (*continued*)

Publication	Year	General study characteristics							LCA characteristics					
		CO_2 source	Capacity, Mt./year	Capture rate,	Capture scheme	System boundary	Unit processes	Temporal scope	Reference Flow	Energy scenarios	Sensitivity	Background modelling	Impact method	Midpoint categories
Bisinella et al. (2022)	2021	Post-combustion at MSWI plant	N	90 %	MEA	Cradle-to-gate	C;T;S	2025-2055	1 ton of wet MSW	Yes	Yes	ecoinvent 3.7.1	EF 3.0	16
Lerche Raadal and Saur (2022)	2022	Post-combustion at steam production	0.05	90 %	HPC	Cradle-to-grave	C;T;S	2020-2021	1 ton of CO_2 stored	No	Yes	ecoinvent 3.6	EN 15804 + A2 method	3
Facchino et al. (2022)	2022	Post-combustion at NGCC power plant	1.1	90 %	MDEA	Cradle-to-gate	C;S	2020	1 kWh of high-voltage electricity	No	No	ecoinvent 3.8	CML 2001	4
Mio et al. (2023)	2022	Power plants, cement kilns, steel mills, and kraft pulp mills	N	95 %	MDEA/MEEP/ gas-liquid absorption	Cradle-to-gate	C;T;S	N	1 t of steel produced	No	No	ecoinvent 3.8	ReCiPe (H) 2016	14
Huo et al. (2022)	2023	Power plants, cement kilns, steel mills, and kraft pulp mills	5.6-14	N	MEA	N	C	2050	1 kg of captured CO_2	2050	Yes	ecoinvent 3.8	N	2
Boakes et al. (2023)	2023	Post-combustion at MSWI plant	N	85-95 %	MEA	Cradle-to-gate	C;T	N	Utilization of 7957 TJ of energy	No	No	ecoinvent 3.8	Umberto LCA	6
Kev et al. (2023)	2023	Post-combustion at coal fired power plant	1.5	90 %	MEA	Cradle-to-grave	C;T	N	1 ton of captured CO_2	Yes	Yes	ecoinvent 3.6	ReCiPe	18
Gallego Dávila et al. (2023a)	2023	Post-combustion from cement production	1.3	N	MEA	Cradle-to-gate	C;T;S	2050	Production of 1 ton of grey cement clinker	2050	No	ecoinvent 3.6	ReCiPe 2008	17
Nilkar et al. (2023)	2023	Membrane-based capture processes	N	90 %	MEEP	Cradle-to-gate	C	N	1 kg of CO_2 per kg CO_2 avoided by the membrane.	No	No	GREET[a]	N	4
Leonzio et al. (2023)	2023	Post-combustion from cement and steel production	20.3-32.8	90 %	MEA	Cradle-to-gate	C;T;S	N	Production of 76.7 Mt. cement / 66.92 Mt. iron and steel.	No	Yes	ecoinvent 3.6	CML 2001	9
Hospital-Benito et al. (2023)	2023	Post-combustion at H_2 production at SMR plants	N	90 %	MDEA & IL	Cradle-to-gate	C;S	N	1 kg of captured CO_2	No	No	ecoinvent 3.7	ReCiPe (H) 2016	9
Shu et al. (2023a)	2023	Post-combustion for cement and steel plants, and NGCC power plants	33-221	90 %	MEA	Cradle-to-grave	C;T;S	2016-2045	1 ton of CO_2 captured, transported and stored.	Yes	Yes	ecoinvent 3.5 APOS	EF 3.0	16
Varling et al. (2023b)	2023	Combustion and upgrading for biogas production (BECCS)	N	85 %	MEA/MEEP and water scrubber	Cradle-to-grave; no construction	C;T;S	2020-2040	per 1000 Nm^3 biogas treated.	Yes	Yes	EASETECH 3.4.0	EF 3.0	16
Haran et al. (2023)	2023	Post-combustion at coal fired power plant	N	85 %	Calcium looping	Cradle-to-grave	C;T;S	N	1 MWh of net electricity produced	Yes	Yes	N	N	1
Xie et al. (2023)	2023	Post-combustion at direct coal-to-liquid production	N	90 %	N	Cradle-to-grave	C;T,S	N	1 ton of oil produced	No	Yes	N	N	1
Burger et al. (2024b)	2024	Post-combustion at various processes	1.5-5	90 %	AMP-PZ	Cradle-to-grave	C;T;S	N	1 ton of CO_2 stored	No	Yes	ecoinvent 3.9.1	EF 3.0	16

Capture (C); Transport (T); Storage (S); Monitoring (M).

AC: activated carbon. AMP-PZ: 2-amino-2-methyl-1-propanol - piperazine. HCLC: hybrid coal chemical looping. HPC: hot potassium carbonate. IL: ionic liquid-based [Bmim][Acetate]. MEA: monoethanolamine. MEEP: membrane-based capture processes. MDEA: methyl diethanolamine. MSWI: municipal solid waste incineration. NGCC: natural gasification combined cycle. VPSA: vacuum pressure swing adsorption.

[a] United States Department of Energy (US DOE) developed Greenhouse gases, Regulated Emissions and Energy in Transition model and database.

[b] China Life Cycle Database.

published between January 2013 and May 2024.

Table 3 summarizes the general study and LCA characteristics. The studies cover a wide range of CCS scenarios based on very different assumptions regarding capture technology, industrial application, the level of inclusion of core CCS activities e.g., means of transport and distances, as well as choices made for background modelling and impact methods. The following presents a breakdown of the general study and LCA characteristics across each study.

4.1.1. CO_2 source, capture capacity, rates and schemes

Most studies focus on post-combustion CCS ($n = 28$) and typically from waste treatment processes, cement and steel production or power generation (namely a coal-fired or natural gas power plant). Two studies focus on CCS at biogenic point sources. Twenty studies do not report the total capture capacity. Those studies that do report the capture capacity show significant variance in the annual capacities, ranging between 9 and 550 Mt. CO_2. Studies published between 2014 and 2020 tend to assume a capture rate ranging from 60 to 99 % whereas later studies assume 85 to 95 %. We also noticed an increased tendency to assume 90 % capture rate without further justification. Monoethanolamine (MEA) capture schemes are the dominant choice for post-combustion amine-based absorption ($n = 22$) typically chosen due to its mature technology readiness level and lower cost compared to other amine solvents.

4.1.2. System boundary and unit processes

Six studies assess the capturing process (C), four studies either capture and transport or storage (T; S), and 23 studies defined a system boundary which included capture, transportation and storage (C; T; S). Generally, information is scarcely provided on technical specifications and data in their life cycle inventories on processes related to interim storage, permanent storage, and preparatory activities for the offshore storage site e.g., well drilling, CO_2 offloading and injection systems and offshore platform modification. Of those studies that include transport and storage processes, only six studies mention interim storage activities with none reporting on material and energy consumption.

4.1.3. Temporal scope and reference flow

Twenty studies state the temporal scope, typically with a timeframe between 2020 and either 2030 or 2050. Most studies use an energy ($n = 13$) or mass-based ($n = 18$) reference flow i.e., per ton or kWh, and generally, we observe a slight tendency to apply a mass-based unit for CO_2 handling, i.e., per ton of CO_2 captured, transported and stored. A mass-based unit seems reasonable for the capture process and comparative studies (CCS vs non-CCS) to account for the energy penalty from introducing carbon capture (CC) into a power generation system. Some papers use two stages in determining GHG emissions; first from compensated electricity production during capturing, and then the life cycle emissions derived from the construction and operation of infrastructure and CO_2 injection (Bisinella et al., 2021; Yujia et al., 2014). This seems prudent since no products are produced from carbon transport and storage processes.

4.1.4. Sensitivity

Sensitivity is an essential part of the final interpretation and is recommended by the ISO standard for any LCA (International Organization for Standardization, 2006a; International Organization for Standardization, 2006b; Laurent et al., 2020). Twenty studies include sensitivity analysis yet poorly defining the sensitivity parameters checked. As many of the reviewed studies depict cases that are based on preliminary, conceptual CCS technologies and designs, testing the sensitivity of the assumptions made is deemed critical to validate the reported results. Similar conclusions were drawn by Terlouw et al. (2021).

4.1.5. Energy scenarios

Modelling of energy scenarios i.e., variations of energy sources, or projection of the electricity carbon intensity i.e., variation in electricity sources or grid mixes, are observed in less than half of the publications ($n = 8$) and typically through projection of the carbon intensity for electricity production for capture and compression processes. Where reported, studies often use a Danish grid mix projection (Karlsson et al., 2016) as the most-optimistic scenarios. See, e.g. Varling et al. (2023a); Bisinella et al. (2021); Bisinella et al. (2022); Shu et al. (2023a). Alternatively, *Premise* was used for prospective LCA projection (Gallego Dávila et al., 2023a; Sacchi et al., 2022) or with reference to the study of Deutz and Bardow (2020).

4.1.6. Background modelling and impact calculation methods

Most studies ($n = 26$) specified which database was used for background modelling (with ecoinvent (Ecoinvent Centre, 2024) most frequently applied) and the impact calculation method ($n = 28$) chosen with ReCiPe (2016), EF. 3.0 and CML 2001 commonly applied. Sensitivity of the impact calculation method was not included in any of the studies although the choice of method may influence the final LCA results.

4.1.7. Environmental impact categories

Table 4 shows the reported midpoint categories. Most studies ($n = 27$) did not state a reference behind the selection of impact categories. When stated, EC-JRC 2011 (Wolf et al., 2010) was used as a reference for the selection and prioritization of mid-point impact categories. Twenty-seven studies include multiple impact categories, yet the number included differs significantly. All reviewed studies reported for the climate change impact category, of which five studies reported only for this category. Prioritization of the climate change impact category was most often justified by the deployment of CCS being driven by climate change reduction. Acidification, freshwater eutrophication, ozone depletion and terrestrial eutrophication are among the commonly included categories besides climate change. Water use and land use are among the categories least reported. In some cases, an increase in water ($n = 6$) and land use ($n = 6$) is reported, whereas acidification and freshwater eutrophication are equally reported as increasing or decreasing. Considering the high level of water consumption for amine-based capture processes, we recommend that water usage is prioritized as a mid-point category for CCS systems. See, e.g. Gallego Dávila et al. (2023a); Burger et al. (2024b); Zakuciová et al. (2020). Generally, for ozone depletion ($n = 8$) and abiotic depletion ($n = 8$), human toxicity ($n = 11$), marine eutrophication ($n = 9$), particulate matter ($n = 9$) and terrestrial eutrophication ($n = 8$) an increase is reported, whereas acidification ($n = 10$) showed a decreased impact in scenarios with CCS. An increase is generally reported for the remaining categories (acidification potential, freshwater ecotoxicity and freshwater eutrophication). Despite the significant energy demanded required for CCS systems, the Cumulative Energy Demand (CED) was only included in three studies by Iribarren et al. (2013), Cuéllar-Franca and Azapagic (2015), Antonini et al. (2020) and Lerche Raadal and Saur (2022). Notably, all 33 studies report a positive contribution to climate change except for two that do not provide any conclusion (Chelvam and Hanafiah, 2023; Morimoto et al., 2023).

Table 4
Impact category coverage for the reviewed articles including trade-off indication based on the reported results. Reported (**R**) but not normalized results reflected for each study. Where normalized and compared to a non-CCS scenario, ↑ reflects an increase within an impact category, and a decrease by ↓ whereas ↕ reflects a mixed outcome depending on the case study scenario reported for.

Publication	Climate change	Water use	Land use	Acidification	Freshwater eutrophication	Ozone depletion	Abiotic depletion	Particulate matter	Freshwater ecotoxicity	Human toxicity	Marine eutrophication/ ecotoxicity	Terrestrial eutrophication/ ecotoxicity
Volkart et al. (2013a)	↓		↑	↑	↑			↑	↑	↑		
Iribarren et al. (2013)	↓			↓		↑	↑	↓			↑	
Yujia et al. (2014)	↓											
Lacy et al. (2015)	↓											
Troy et al. (2016)	↓			↑	↑		↑	↑	↑	↑	↑	↑
Štefanica et al. (2016)	↓											
Navajas et al. (2019)	↓		↑	↕	↑	↑	**R**	↑	↑	↑	↕	↕
Young et al. (2019)	↓	↑		↓	↑			↑				
Thorne et al. (2019)	↓			↓				↕		↑		
An et al. (2019)	↓			↓	↓	↑	↑		↑	↑	↑	↑
Kim et al. (2019)	↓											
Yang et al. (2019)	↓			↓	↑	↑	↓	↑	↑	↑	↑	↑
Zakrisson et al. (2023)	↓	↑	↑	↓	↓		**R**	↓	↑	**R**	**R**	↓
Zakrisson et al. (2023)	**R**	**R**	**R**	**R**	**R**	**R**	**R**	**R**	**R**	**R**	**R**	**R**
Bello et al. (2020)	↓	↑	↑	↑	↑	↑	↓		↑	↑	↑	
Antonini et al. (2020)[b]	↓		**R**	**R**	**R**	**R**	**R**	**R**	**R**	**R**	**R**	**R**
Bisinella et al. (2021)	↓	**R**		↓	↓	↓	↓	↓	↓	↓	↑	↓
Bisinella et al. (2022)[b]	↓	**R**	↓	↓	↓	↓	↓	↓	↑	↓	↓	↓
Lerche Raadal and Saur (2022)	↓		↑	↑								
Facchino et al. (2022)[b]	↓			↑	↑	↑	↑					
Mio et al. (2023)	↓	↕		↕	↑		↕	↕	↑	↑	↕	↑
Huo et al. (2022)	↓							**R**				
Boakes et al. (2023)[b]	↓			↓	↓							↓
Kev et al. (2023)	↓	**R**			**R**	**R**		**R**	**R**		**R**	**R**
Gallego Dávila et al. (2023a)	↓	↑	↕	↑	↕	↕	↕	↑	↑	↑	↑	↕
Nilkar et al. (2023)	**R**			**R**			**R**					
Leonzio et al. (2023)[a]	↓			↑	↑	↑	↑		↑	↑	↑	↑
Hospital-Benito et al. (2023)	↓			**R**	**R**	**R**	**R**		**R**	**R**	**R**	**R**
Shu et al. (2023a)	↕	↑	↑	↑	↑	↓	↑	↑	↑	↑	↑	↑
Varling et al. (2023b)	↓	**R**	**R**	**R**	**R**	**R**	**R**	**R**	**R**	**R**	**R**	**R**
Haran et al. (2023)	↓											
Xie et al. (2023)	↓											
Burger et al. (2024b)[b]	↓	**R**	**R**	**R**	**R**	**R**	**R**	**R**	**R**	**R**	**R**	**R**

[a] Impacts not normalized but presented as a MDEA - IL comparison.
[b] European Commission's Recommendation on the use of Environmental Footprint methods, 2021 (Wolf et al., 2010).

4.2. Life cycle inventories

Table 5 presents the life cycle inventory data for each CCS process across the reviewed LCAs. Generally, larger variances were found in the reported input and output values which challenge comparison of the reviewed studies. Particularly, inconsistency is observed within the general study and LCA characteristics, hereunder, boundary incompleteness, definition of the functional unit and modelling choices as well as the overall goal and scope. Instead, an attempt was made to assess the accuracy of the inventories by comparing the reported impacts with calculated impacts based on theoretical values (see also Section 3.4).

4.2.1. Capturing and compression

All studies except one provide energy consumption data for capture and compression processes. However, the reported energy demand significantly differs across the studies, ranging from 0.15 to 3750 kWh/tCO_2. The large difference is likely due to heat integration, which is often seen in waste or energy plants, and to some extent in industrial processes such as cement production. However, this was only indicated in a few cases e.g. Bisinella et al. (2021); Bisinella et al. (2022). Additionally, it is not always clear whether energy consumption for compression is included or not in the data for energy consumption for capturing. Depending on how CO_2 is transported, the compression stages change. For instance, it is often appropriate to transport CO_2 via pipelines in the gas phase, whereas it will be advantageous to liquefy the CO_2 in liquid form for interim storage or transport in tanks. The reported consumption of MEA solvent ranges from 0.002 to 2.24 kg/tCO_2 captured without further justification or explanation to the consumption levels. Material use for capture and liquefaction facilities were reported in two studies (Troy et al., 2016; Shu et al., 2023a) anticipating a total steel demand ranging from 789 to 2900 ton for the capture and liquefaction plant.

4.2.2. Transport

In 19 cases, data is either provided for a single mode of transport, or the study provides a consolidated input for all transport stages across the entire value chain, whereas eight studies report more than one mode of transport. Twelve studies envisage pipelines as the only means of transport, and here, the quantification of impacts was approached quite differently across the studies. Shu et al. (2023a) provide one of the most detailed estimations for onshore pipelines although distinctly based on Wildbolz (2007) who assumes a CO_2 recompression of 0.01207 kWh per ton/km with recompression needed for every 200 km. The pipeline design parameters were most often based on Wildbolz (2007). The construction of electrical compressor stations was assumed by using the model for the construction of 10 MW gas turbines in Europe using ecoinvent 3.5 APOS as proxy with transport modelled between nodes as transshipment process in SecMOD framework (Reinert et al., 2022). According to Koornneef et al. (2008) the construction process is given for a pipeline with a capacity of 94 Mt. over a 20-year lifetime will a full load hour of 7800 h. Bisinella et al. (2022) also covers compressor, pipeline transport and construction in relevant details. Based on Knoope et al. (2013), the authors assume that specific booster station needed each 100 km with pressure loss per booster being 5.67 bar and electricity consumption per booster station as 0.26 kWh/tCO_2. Authors also provide pipeline diameter, thickness, steel density and mass of 55,168 t for the pipeline of 850 km. Kev et al. (2023) provides some relevant information on the construction of 1 km of pipeline, such as occupation for construction site, transformation of forest, amount of steel, rock wool, usage of helicopters, based on Aspen simulations with reference to Øi (2012). Fuss et al. (2018) also made some important consideration for quantities of steel, aluminum, polyethylene, gravel and diesel for construction of 1 km of pipeline. In those cases considering an onshore or offshore pipeline connection to the permanent storage site, pipeline lengths between 200 and 1000 km were typically assumed. In some cases, offshore pipelines were modelled as an alternative, typically based on the model of European long-distance transport for natural gas, straightly replacing LNG transport by transport of CO_2. Thus, operating conditions are rarely differentiated between onshore and offshore pipelines. Information on pipeline specifications such as outer diameter, corrosion allowances, steel types and chrome content, although these factors have an impact on material requirements.

None of the studies report on the design parameters or technical specifications for onshore vehicle transportation. For vessel transportation, anticipation of the CO_2 vessel carrier capacities and fuel consumption was reported in one study (Burger et al., 2024b) although several studies report sensitivity related to the transport processes. See e.g. Emborg et al. (2024), Iribarren et al. (2013) and Kim et al. (2019). The energy or fuel requirement for any type of vehicle or vessel was not reported in any of the studies except for Huo et al. (2022) reporting a fuel consumption of 5.41 g fuel per ton of CO_2 transported, and Leonzio et al. (2023) reporting an electricity consumption of 0.011 kWh per t*km although without specifying the technical specifications on sizes and gross tonnage. Cradle-to-grave impacts of vehicles and road or harbor infrastructures were not specified in any of the studies.

4.2.3. Storage and injection

Energy requirements for offshore storage, i.e., injection equipment and platform operations, were included in 12 studies. For instance, Antonini et al. (2020) report values for CO_2 transport and storage based on assumed distances and life cycle inventory data for permanent storage in saline aquifers based on Volkart et al. (2013b). Alternatively, impacts were modelled using assumed capacities of offshore wells by Wildbolz (2007) while using the Ecoinvent process for oil and gas drilling (Cuéllar-Franca and Azapagic, 2015; Hospital-Benito et al., 2023; Shu et al., 2023a). Specific energy consumption data for offshore injections as provided in four cases with estimations ranging from 0.01 to 134 kWh/tCO_2 injected. For injection into onshore reservoirs, two studies reported an energy consumption of 0.01 and 92 kWh/tCO_2 injected. Notably, Volkart et al. (2013a) evaluates the sensitivity related to the distance to the permanent storage site as well as storage depths, showing a potential increase in impacts between 2 and 10 % when increasing pipeline and storage depths from 200 to 400 km and 1000 and 3000 m respectively. Generally, material requirements for constructions are poorly covered with few studies providing inventory data for steel and concrete requirements of the capture facilities and pipelines. See e.g. Bello et al. (2020), Shu et al. (2023a) and Gallego Dávila et al. (2023b). The assumed lifespan of structures and assets typically ranges from 20 to 30 years.

4.2.4. Monitoring and preparatory actions

Six studies included preparatory activities, i.e., well drilling. Monitoring activities related to subsurface plume migration and leakage monitoring were reported in three cases, and in no cases were seismicity included. Drilling of injection wells were reported in six studies for storage depths, i.e., drilled meters to quantify the environmental impact by using an average ecoinvent process for offshore drilling (Antonini et al., 2020). For instance, Hospital-Benito et al. (2023), Shu et al. (2023b) and Antonini et al. (2020) calculates the impacts based on global drilling processes from ecoinvent, assuming well depths between 1000 and 1200 m, and Kev et al. (2023) specified drilling deep borehole up to 3360 m. Generally, details and input values such as material requirements and waste related to well drilling or re-drilling of existing oil and gas wells for CO_2 injection were not provided. Notably, only twelve studies concretize whether the permanent storage location is onshore or offshore. Comparing values for offshore drilling from literature (Emborg and Olsen, 2024) could potentially lead to larger variances in the impact results for well drilling depending on where the permanent storage site is located which will be further discussed in Section 5.

Table 5
Life cycle inventory (LCI) values for technical flows for each process. Not applicable (NA) indicate that the process was not included in the system boundaries. A dash indicates inclusion with no reporting of the numerical input value.

Publication	Capture and compression							Transport										Storage				Monitoring		Preperatory	
	Capture				Compression	Liquefaction	All	Pipeline (onshore)			Truck		Pipeline (offshore)			Vessel		Interim		Onshore	Offshore	Plume	Leak	Seismic	Wells
	Electricity use, kWh/ tCO_2	Steam use, kWh/ tCO_2	Solvent use, kg/ tCO_2	Water use, m^3/ tCO_2 [1]	Electricity use, kWh/ tCO_2	Electricity use, kWh/ tCO_2	Material use, Steel, ton	Electricity use, kWh/ tCO_2	Material use, Steel, ton	Distance, km	Energy use, t/ tCO_2	Distance, km	Electricity use, kWh/ tCO_2	Material use, Steel, ton	Distance, km	Energy use, t/ tCO_2	Distance, km	Electricity use, kWh/ tCO2	Material use, Steel, ton	Electricity use, kWh/ tCO_2	Energy use, kWh/ tCO_2	Energy use, MJ/ year	Energy use, tCO_2/ year	Energy use, MJ/ year	Energy use, MJ/ year
Volkart et al. (2013a)	0.02–0.2	–	0.0003	0.08	–	NA	NA	–	–	200–400	NA	NA	NA	NA	NA	NA	NA		NA			NA	NA	NA	–
Iribarren et al. (2013)	–	NA	0.002 kg/ kWh	NA	0.02–0.03 kWh/RF [2]		NA	0.02–0.03 kWh/RF [2]	–	680	NA	NA	NA	NA	NA	NA	NA	NA	NA	0.02–0.03 kWh/R [2]		NA	NA	NA	NA
Yujia et al. (2014)	–	–	1.5	–	111	NA	NA	–	–	50	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	7	NA	NA	NA	NA
Lacy et al. (2015)	–	NA	–	–	–	NA	NA	NA	NA	NA	NA	NA	–	–	91.5	NA	NA	NA	NA	NA	–	NA	NA	NA	NA
Troy et al. (2016)	–	NA	–	NA	–	NA	2900	–	–	400	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	–	NA	NA	NA	–
Štefanica et al. (2016)	337	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	NA	–	NA	NA	NA	NA
Navajas et al. (2019)	7	NA	0.35–2	NA	5	NA	NA				0.002 $kgCO_{2eq}$/$kgCO_2$								NA			NA	NA	NA	NA
Young et al. (2019)	0.7–1.1	–	–	–	0.1	NA	NA	–	–	588	NA	NA	NA	NA	NA	NA	NA	NA	NA	–	–	NA	NA	NA	NA
Thorne et al. (2019)	23–175 kWh	NA	0.003 kg/ kWh	NA	103	NA	NA					NA							NA			NA	NA	NA	NA
An et al. (2019)	23–175	–	2.24	–	–	–	–	–	–	–	–	–	–	4000	200	–	–	–	–	NA	668	NA	NA	NA	NA
Kim et al. (2019)	0.1	NA	NA	0.001	NA	NA	NA	141	348	100	NA	NA	NA	NA	NA	NA	NA	NA	NA	26	NA	NA	NA	NA	–
Yang et al. (2019)	600 MW	–	–	–	111	NA	–	2.4	–	200	NA	NA	NA	NA	NA	NA	NA	NA	NA	7	NA	NA	NA	NA	NA
Zakuciová et al. (2020)	23 MW	NA	NA	9259 ton/ h	NA	NA	NA					NA							NA			NA	NA	NA	NA
Zakuciová et al. (2020)	0.01	NA	NA	9259 ton/ h	NA	NA	NA					NA							NA			NA	NA	NA	NA
Bello et al. (2020)	0.2 kWh/ RF	–	0.001 kg/ RF	2.7 kg/RF	40 kWh/RF	–	–	–	270	200	–	–	–	–	–	–	–	NA	NA	NA	–	NA	NA	NA	NA
Antonini et al. (2020)	–	–	–	NA	NA	NA	NA			0.01 kWh / $kgCO_2$ transported across 200 km pipeline and stored								NA	NA	NA	–	NA	NA	NA	–
Bisinella et al. (2021)	15	1027	4	NA	65–80	NA	NA				0.02 $kgCO_{2eq}$ / tCO_2 across 1000 km								2% overall losses			NA	NA	NA	NA
Bisinella et al. (2022)	42	2.5	1	NA	105	42	NA				0.02 $kgCO_{2eq}$ / tCO_2 across 1150 km								2% overall losses			NA	NA	NA	NA
Lerche Raadal and Saur (2022)	278	NA	NA	NA	–	83	–	–	–	–	–	38	–		700	–	–		–	NA	–	NA	NA	NA	NA
Facchino et al. (2022)	933	1087	2	0.4	NA	NA	NA					NA						NA	NA	92	NA	NA	NA	NA	NA

(*continued on next page*)

Table 5 (*continued*)

Publication	Capture and compression							Transport										Storage				Monitoring		Preperatory	
	Capture				Compression	Liquefaction	All	Pipeline (onshore)			Truck		Pipeline (offshore)			Vessel		Interim		Onshore	Offshore	Plume	Leak	Seismic	Wells
	Electricity use, kWh/ tCO_2	Steam use, kWh/ tCO_2	Solvent use, kg/ tCO_2	Water use, m^3/ tCO_2 [1]	Electricity use, kWh/ tCO_2	Electricity use, kWh/ tCO_2	Material use, Steel, ton	Electricity use, kWh/ tCO_2	Material use, Steel, ton	Distance, km	Energy use, t/ tCO_2	Distance, km	Electricity use, kWh/ tCO_2	Material use, Steel, ton	Distance, km	Energy use, t/ tCO_2	Distance, km	Electricity use, kWh/ tCO2	Material use, Steel, ton	Electricity use, kWh/ tCO_2	Energy use, kWh/ tCO_2	Energy use, MJ/ year	Energy use, tCO_2/ year	Energy use, MJ/ year	Energy use, MJ/ year
Mio et al. (2023)	–	–	0.1 kg/ RF	211 kg/h	–	–	–	–	–	–	–	–	–	–	–	–	–		NA			NA	NA	NA	NA
Huo et al. (2022)	0.2	1.1	0.002	NA	–	NA	NA	5.4 g fuel / tCO_2 across 470 km											NA			NA	NA	NA	NA
Boakes et al. (2023)	14	0.888	0.8	30	NA	101	NA	NA	–	NA	NA	NA	NA	NA	NA	0.002	470	–	-	NA	–	NA	NA	NA	NA
Kev et al. (2023)	3360-3740	–	–	NA	90	–	–	2.38	55,168	850	NA	NA	NA	NA	NA	NA	NA	–	–	NA	–	NA	NA	NA	NA
Gallego Dávila et al. (2023a)	42	267	0.58	0.1	–	138	–	NA	NA	NA	NA	NA	–	–	–	193	480–730	–	–	NA	134	NA	NA	NA	NA
	733	305						NA	NA	NA	NA	NA								NA		NA	NA	NA	NA
	304	201						NA	NA	NA	NA	NA								NA		NA	NA	NA	NA
Nilkar et al. (2023)	–	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA		NA			NA	NA	NA	NA
Leonzio et al. (2023)	236	–	–	–	–	NA	NA	400 km and 0.011 kWh/t*km with construction included											NA			NA	NA	NA	NA
Hospital-Benito et al. (2023)	0.15–0.2	0.25–0.3	0.002	0.1	–	NA	NA	0.03	–	–	NA	NA	NA	NA	NA	NA	NA	NA	NA	0.01	NA	NA	NA	NA	–
Shu et al. (2023a)	130–429	429–960	1.2–1.9	–	5–45	NA	789	0.0792 kWh / tCO_2 transported and stored by onshore and offshore pipelines at a fixed distance of 1000 km (including recompression every 200 km and construction impacts)												NA			NA	NA	NA	–
Varling et al. (2023a)	28–200	600–1893	3–7	0.33–0.40	105	NA	NA	NA										NA	NA	NA	–	NA	NA	NA	NA
Haran et al. (2023)	–	NA	–	–	–	NA	NA	- [3][4]	–	200	NA	NA	NA	NA	NA	NA	NA	NA	NA	-[4]		NA	NA	NA	NA
Xie et al. (2023)	940	NA	NA	NA	NA	NA	NA	1.3	–	–	NA	NA	NA	NA	NA	NA	NA	16	–	NA	NA	NA	NA	NA	NA
Burger et al. (2024a)	17	–	0.4	0.2	NA	97	NA	NA	NA	NA	0.0002	4304	–	–	–	0.00003	800	–	–	NA	–	NA	NA	NA	NA

NA: process not included in system boundaries.

-: process included but numerical values not reported in the life cycle inventory.

[1] Water consumption reflects the total requirement, incl. deionized water (make-up) and cooling water.

[2] Volkart et al. (2013b) reported electricity requirements for compression (to liquid phase), pipeline transport (680 km) and injection into geological formations.

[3] Design parameters for pipeline transport are based on Wildbolz (2007).

[4] Energy requirement for pipeline transport and injection into geological storage reservoirs is based on Koornneef et al. (2008).

Table 6
Summary of the reported LCIA outputs measured in climate change impact (kgCO2eq). The level of study completeness in pattern-indicated as follows: # = the highest level of inclusion. + = medium level of inclusion. * = lowest level of inclusion.

Publication	Reference flow	$kgCO_{2eq}$/reference flow (RF)									CCS value chain efficiency			Electricity carbon intensity			Study completeness			
		Capture and Compression	Solvent	Liquefaction	Transport, pipeline	Transport, truck	Transport, vessel	Injection & Storage	Monitoring	Preparatory	Captured CO2 [MT/y]	Value chain emissions [tCO_{2eq}/RF]	Net-stored [%]	Baseline [$kgCO_{2eq}$/kWh]	Projected [$kgCO_{2eq}$/kWh]	Change [%]	Solvent losses	Process coverage	Granularity	Impact category
Volkart et al. (2013a)	1 kWh / 1 kg cement	Not reported at a process-level									–	–	39–92%	NA	NA	NA	Yes	+	*	*
Iribarren et al. (2013)	1 kWh of net electricity	Not reported at a process-level									–	–	56–73%	NA	NA	NA	Yes	+	*	*
Yujia et al. (2014)	1 MWh produced	97–131	–	NA	1.8–2.1	NA	NA	7.1–8.2	NA	NA	0.8	139–183	–	NA	NA	NA	No	+	*	*
Lacy et al. (2015)	1 kWh produced	0.177	–	NA	–	NA	NA	–	–	NA	1	–	21–36%	NA	NA	NA	No	+	+	+
Troy et al. (2016)	1 kWh produced	–	–	NA	–	NA	NA	–	NA	NA	–	–	74–77%	NA	NA	NA	No	+	*	#
Štefanica et al. (2016)	Production of 1 kWh of electrical energy	0.53	–	NA	–	NA	NA	–	–	–	1.2	0.53	0.62	NA	NA	NA	No	*	+	*
Navajas et al. (n.d.)	Production of 1 MWh	Not reported at a process-level									–	–		NA	NA	NA	No	*	*	#
Young et al. (2019)	1 kg of CO_2 equivalent abated	0.7–1.4	–	NA	–	NA	NA	–	NA	NA	1205	0.30–0.36	62–82%	NA	NA	NA	No	+	*	+
Thorne et al. (2019)	1 kWh of net electricity	–	–	NA	NA	NA	NA	NA	NA	NA	–	700	–	NA	NA	NA	No	*	*	+
An et al. (2019)	1 ton of cement	Not reported at a process-level									–	–		NA	NA	NA	No	+	+	#
Kim et al. (2019)	0.7 Mt. CO_2 captured per year	0.2	NA	NA	0.1	NA	NA	0.01	NA	–	0.2	3.2	0.64	NA	NA	NA	No	+	+	+
Yang et al. (2019)	1 MWh of net power produced	–	–	NA	NA	NA	NA	NA	NA	NA	–	–	62–77%	NA	NA	NA	No	+	+	#
Zakuciová et al. (2020)	Electricity production by 250 MW coal power unit	–	–	NA	NA	NA	NA	NA	NA	NA	–	0.04–0.3	–	NA	NA	NA	No	*	*	#
Zakuciová et al. (2020)	1 kWh of produced electricity	–	–	NA	NA	NA	NA	NA	NA	NA	–	–	–	NA	NA	NA	No	*	*	#
Bello et al. (2020)	1 km of distance travelled	0.04–0.06			Not reported at a process-level						–	–		NA	NA	NA	No	+	+	
Antonini et al. (2020)	1 MJ of compressed H_2	–	–	NA	–	NA	NA	–	NA	NA	NA	0.003–0.006	NA	NA	NA	NA	No	+	+	#
Bisinella et al. (2021)	1 ton wet MSW	17	12	–	29			–	–		0.3–1.0	0.06	94–98%	0.21	0.01 [3]	0.05	No	+	+	#
Bisinella et al. (2022)	1 ton wet MSW	49.6–62.1	3.53	NA	39			NA	NA		NA	0.04	0.95	0.17	0.02 [3]	0.21	No	+	+	#
Lerche Raadal and Saur (2022)	1 ton CO_2 stored	–	–	NA	–	NA	–	–	–	–	0.05	0.03–0.13	87–97%	NA	NA	NA	No	+	+	+
Facchino et al. (2022)	1 kWh of high-voltage electricity	0.02–0.08	–	NA	NA	NA	NA	NA	NA	NA	1.1	0.00002–0.00008	89–97%	NA	NA	NA	No	*	*	+
Mio et al. (2023)	1 ton of steel produced	–	–	–	–	–	–	–	–	–	11	0.002–0.014	90–95%	NA	NA	NA	Yes [1]	+	+	#
Huo et al. (2022)	1 kg of captured CO_2 compressed	0.6–0.9	–	NA	–	–	–	–	NA	NA	5.6–14	–	–	–	–	–	Yes [1]	*	*	+
Boakes et al. (2023)	Utilization of 7957 TJ of energy	Not reported at a process-level									–	–	0.9	NA	NA	NA	No	*	*	+
Kev et al. (2023)	1 ton CO_2 captured	275–282	103	NA	–	NA	NA	NA	NA	NA	0.001	–	27–36%	–	–	–	No	+	+	#
Gallego Dávila et al. (2023a)	Production of 1 ton of grey cement clinker	80–100		NA	NA	NA		30–50			1.3	–	–	0.28	0.1 [2]	–	No	+	+	#
Nilkar et al. (2023)	1 kg of CO_2 per kg CO_2 avoided	0.00004–0.008		NA	NA	NA	NA	NA	NA	NA	–	–	–	NA	NA	NA	No	*	*	+

(*continued on next page*)

Table 6 (*continued*)

Publication	Reference flow	$kgCO_{2eq}$/reference flow (RF)									CCS value chain efficiency			Electricity carbon intensity			Study completeness			
		Capture and Compression	Solvent	Liquefaction	Transport, pipeline	Transport, truck	Transport, vessel	Injection & Storage	Monitoring	Preparatory	Captured CO2 [MT/y]	Value chain emissions [tCO_{2eq}/RF]	Net-stored [%]	Baseline [$kgCO_{2eq}$/kWh]	Projected [$kgCO_{2eq}$/kWh]	Change [%]	Solvent losses	Process coverage	Granularity	Impact category
Leonzio et al. (2023)	Production of 76.7 Mt. cement / 66.92 Mt. iron and steel	–	–		Not reported at a process-level				NA	NA	–	–	–	NA	NA	NA	No	*	*	#
Hospital-Benito et al. (2023)	1 kg CO_2 captured	0.16–0.17	0.005	NA	0.002	NA	NA	0.004	NA	0.0001	NA	0.0002	NA	NA	NA	NA	No	+	+	#
Shu et al. (2023a)	1 ton of CO_2 captured, transported and stored	–	–	NA	–	NA	NA	–	NA	–	33–221	–	88–95%	–	[3]	0.3	No	+	+	#
Varling et al. (2023a)	per 1000 Nm^3 biogas treated	–	–	NA	NA	NA	NA	–	NA	NA	–	–	–	0.16	1.06–0.02 [3]	–	No	*	*	#
Haran et al. (2023)	1 MWh of net electricity produced	–	NA	NA	–	NA	NA	–	NA	NA	–	0.149	79–83%	0.72	0.7–0.83 [4]	–	No	*	+	+
Xie et al. (2023)	1 ton of oil produced	–	NA	NA	–	NA	NA	–	NA	NA	–	0.52	28–57%	NA	NA	NA	No	*	+	
Burger et al. (2024a)	1 ton of CO_2 stored	100–260	–	NA		80–150		5	NA	NA	–	0.3–0.5	50–70%	–	–	–	No	+	+	#

NA: process not included in LCIA.

-: process included in LCIA but result not provided.

Study completeness indicators: Process coverage (* =C;T, + =C;T;S, # =C;T;S;M). Process granularity (* =c;e, +l=c;e;m;w, # =c;e;m;w). Impact category coverage: * = 1 or 2 midpoints, + = > two midpoints across more than one endpoint, # = > five midpoints across more than one endpoint. Note: Capture (C); Transport (T); Storage (S); Monitoring (M); Chemicals (c); Energy (e); Materials (m); Waste (w).

[1] MEA degradation considered.

[2] Sacchi et al. (2022) and ENTSO-E (2020).

[3] Karlsson et al. (2016).

[4] CO_2 baseline database for the power sector in India, CEA 2002a.

4.3. Life cycle assessment output

Table 6 summarizes the reported outputs of the life cycle impact assessment (LCIA) for each study. Notably, the units are not harmonized due to differences in the general study and LCA characteristics. Below specifies the major variances observed across the reviewed studies that underpin incoherence across reported results and study completeness and which will be brought up for discussion in Section 5.

4.3.1. Climate change

Capturing and compression processes seemingly account for the largest overall impact contribution yielded by electricity consumption. All studies lack reporting of LCIA outputs to some extent across the processes said included in the assessed value chain. While reporting of climate change impact is consistently provided for the capture process, only seven studies report the output for transport and storage processes. In four studies, the total output consolidating all CCS processes is reported.

4.3.2. Value chain efficiency

Nineteen studies report on value chain efficiency and the total amount of CO_2 stored i.e., the net-stored volume minus the emissions from life cycle activities and leakages. Overall, the studies report an efficiency ranging from 27 to 98 % of the captured amount, anticipating a capture rate between 85 and 95 %. We see a similar conclusion made by Leonzio et al. (2023) whom exemplify the difference across a few LCA studies of CCS supply chains and show deviations ranging from 39 to 82 % across climate change impacts depending on the CO_2 source and capture technology.

4.3.3. Electricity projection

Five out of nine studies reported the electricity carbon intensity factors used for the baseline and projections. The projected factors range from 0.01 to 0.09 $kgCO_{2eq}$/kWh in a low scenario, leading to an overall reduction in the total impact between 5 and 30 %. The baseline electricity carbon intensity factors used range between 1.1 $kgCO_{2eq}$/kWh (electricity based on coal and 0.09 $kgCO_{2eq}$/MJ based on oil) to non-fossil-based energy systems ranging from 0.021 $kgCO_{2eq}$/kWh (electricity based on offshore-wind power) and 0.09 $kgCO_{2eq}$/MJ (based on electricity driven heat pumps). Different approaches for electricity projection are also observed e.g. Gallego Dávila et al. (2023a) whom apply the *Premise* (Sacchi et al., 2022) modelling tool in their prospective LCA.

4.3.4. Study completeness

We established a set of criteria to help evaluate the study completeness of the reviewed publications i.e., to what extent does each study include CCS processes which are considered compulsory to permanently store CO_2, e.g., injection and monitoring. Since CCS is a new technology with the first projects being developed and formal regulation pending, the aim was to try to uncover if there is a tendency to include or exclude certain CCS processes in the LCAs, e.g., due to data availability or a general lack of knowledge. Two key criteria were considered: the level of process coverage i.e., the inclusion of CCS processes throughout the entire value chain, and the types of input flows included for each process i.e. process granularity. The * sign indicates the lowest level of completeness, the + sign a medium level and # the highest level. For process coverage and granularity, the following scores were used to determine the sign indicator respectively: * (capture, transport; chemicals and/or energy), + (capture, transport and/or storage and/or monitoring; chemicals, energy and/or materials, and/or waste), and # (capture, transport, storage and/or monitoring; chemicals, energy, materials and/or waste). For impact category coverage, the following scopes were used to determine the sign indicator: * (<2 midpoints), + (>2 mid-points across one endpoint), and # (>5 mid-points across more than one endpoint).

Generally, concise and consistent reporting of CCS processes (i.e. coverage level) and input flow (i.e., granularity level) is lacking. See Table 6. Twenty studies cover the capture, transport and storage processes, where eighteen of these studies include energy and materials demand inputs only. Fifteen studies report just the energy demands for capture and/or transport processes. Material demands for infrastructural activities were reported in four studies. Material demands and generated waste from intermediate storage, monitoring and preparatory activities such as well drilling or modification of offshore platforms, or the construction of interim storage tanks and foundations, were not covered in a single study. In terms of impact category coverage, 17 studies include >5 mid-point categories across more than one endpoint while the remaining thirteen studies cover between 5 and 1 midpoints for one endpoint. Twelve studies report on >10 midpoint categories whereas three studies report in coherence with the sixteen common LCA impact categories as outlined by the European Commission (Directorate-General for Environment, 2021).

4.4. Validation

Uncertainty in the reported LCA input values for energy demand were tested by applying mechanistic modelling as explained in Section 3.4. Table 7 presents the observed variances between reported energy demands (presented in Table 6) when compared to theoretical, practical exergy demands for the tested CCS processes: capture, compression, liquefaction, pipeline, truck transport and reservoir injection. Formulas

Table 7
Observed variances between reported energy demands and theoretical optimum values for tested CCS processes.[1]

Processes	Reported values	Theoretical values	Technical assumptions
Capture	Lower and higher values reported.	Blower: 43 kWh/ton CO_2 Circulation pump: 3 kWh/ton CO_2	95 % capture rate. Reboiler heat duty = 3080 MJ/ton CO_2 Heat loss by the hot gas streams: 231 MJ/ton CO_2
Compression	17 studies report lower values. Rest is within range.	108 kWh/ton CO_2 (387 MJ/ton CO_2) electricity	Two-stage compression from 1 to 80 bar. Compressor: mechanical efficiency of 80 % and an electrical driver efficiency of 90 %
Liquefaction	Lower values reported.	144 to 156 kWh/ton CO_2 (519 to 562 MJ/ton CO_2)	COP[1] of 1.2 CO_2 pressure/temp.: 6-7 bar / -5.0 to 2.7 °C
Pipeline	Not applicable	70 MJ/ton CO_2 [2]	Mass of carbon steel: 66 to 246 ton/km Pressure drops per unit length: 0.1 bar/km. Maximum pressure: 130 bar. Five pressure boosters (pumps).
Truck transport	Not applicable	356 MJ/ton CO_2 for transporting over 1000 km (15.2 MJ/km or 37.5 L diesel/100 km)	40 % efficiency. Truck weight and gross capacity: 15 tons carrying 40 m^3 of liquid CO_2 at 15 bars (42.8-ton CO_2)
Injection (supercritical CO_2)	Not applicable	Via pipeline: 12.2 MJ electricity per ton CO_2 Via platform: 40.8 MJ fuel ton CO_2.	Injection pressure = 121.5 bar for the injection of 1 Mt. CO_2. Fuel: diesel or gas.

[1] COP: coefficient of performance.
[2] Excluding 387 MJ/ton CO_2 for the initial compression.

and calculations of the theoretical values along technical assumptions are provided in **SM 2**. The following summarizes key findings from the comparison.

4.4.1. Capture

Eight studies report the electrical demand for capture by MEA or MDEA and at capture rates between 85 and 90 %. The reported values range between 0.15 and 933 kWh/ton CO_2 captured compared to the theoretical value of 46 kWh/ton CO_2. For instance, Bisinella et al. (2021) and Bisinella et al. (2022) report electricity uses between 15 and 42 kWh/ton CO_2 with heat integration and thus within the range of theoretical values. Facchino et al. (2022) report an electricity use of 933 kWh/ton CO_2 which is significantly higher than the theoretical value. Hospital-Benito et al. (2023) report lower demands between 0.15 and 0.18 kWh/ton CO_2. While key details on the equipment and technical specifications are lacking in the studies, it is challenging to point out a single reason why the reported values are either lower or higher than the theoretical estimated ones. The rest of the studies, electricity demands are generally not distinguished between electricity use, heat generation, compression and liquefaction. Notably, none of the reviewed studies report on the energy demand for circulating liquid solvents or the blower for transport of CO_2 through the system.

4.4.2. Compression

Ten studies report on the electricity demand for compression of CO_2. The reported values range between 0.1 and 111 kWh/ton of CO_2, hereof six studies report values that are lower than the theoretical value of 108 kWh/ton CO_2. The rest of the studies report values that are within range of the theoretical value between 105 and 111 kWh/ton CO_2. Generally, the studies do not specify the mechanical and electrical driver efficiency nor the number of compression stages, temperature and pressure ranges or the use of multistage pumps.

4.4.3. Liquefaction

Five studies report the electricity demand for liquefaction of CO_2 which ranges between 42 and 138 kWh/ton CO_2 and thus lower than the theoretical value between 144 and 156 kWh/ton CO_2. The variance may be linked to the compressor type and efficiency as well as temperature and pressure differences which were generally not specified in the studies. Only four studies report the energy demand for onshore vehicle transport, generally without specifying the vehicle weight, size or gross tonnage or the energy or fuel type. Thus, the reported values are deemed insufficient for comparison of theoretical values for truck transportation, which was also the case for the comparison of pipeline and injection. Two studies reported the demand for pipeline transportation although with limited insight about steel thickness, material type and pump specifications, and with only Kev et al. (2023) providing the material amount and distance. Regarding injection of CO_2 into geological formations, three studies provided energy consumption for permanent storage activities. Generally, a lack of granularity in current reporting is observed across all CCS processes with details lacking on equipment specific energy consumptions and technical specification.

5. Discussion

In this Section, the results of the literature review are evaluated in Section 5.1. A list of recommendations provided in Section 5.2 along with a critical discussion of the study limitations, literature research protocol and further work required presented in Section 5.3.

5.1. Evaluation

Generally, comparability of the LCAs is challenged by the variety in the general study characteristics. For this reason, the discussion will be centered on areas that seem to be consistently neglected from a reporting perspective. Since each LCA has in common that they assess a CCS system, we chose to evaluate each study based on their inclusion or neglection of processes to be considered an integral part of any CCS value chain, namely, capturing, transport, injection, storage, monitoring and drilling.

5.1.1. General study characteristics and completeness

We found significant variations in the CCS system boundaries and the processes included in the 33 reviewed LCA case studies. Generally, our conclusions are aligned with findings from Chelvam and Hanafiah (2023) that most studies tend to include a cradle-to-gate approach accompanied by a study objective of understanding full life-cycle environmental impacts. Few studies reported on impacts related to CO_2 transportation as well as interim and permanent storage processes which are generally excluded from the boundaries. A description of the applied technologies, capacities and specifications applied after the point of capturing is also lacking. A similar observation was made by Terlouw et al. (2021), stating that the stages of carbon transporting and storing were neglected which increases the chance of excluding possible environmental impacts. In the broader literature, only a few examples were found that more narrowly address the environmental impacts related to CO_2 transportation and storage processes. Here, one study was found conducting techno-economic assessment explicitly of carbon transport infrastructures via pipeline (Zanobetti et al., 2023), and one review study focused on subsurface leakages and monitoring technologies (Roberts and Stalker, 2020). Burger et al. (2024b) showed the importance of leakage during transportation and interim storage by performing sensitivity analysis.

Regarding transportation, An et al. (2019) reported that for cement production post-combustion capture the environmental impact of transportation and storage was minimal, regardless of the carbon capture technology. In general, few studies tend to focus on the impacts related to transportation nor the possibility of a rebound effect from upscaling CCS value chains and thus increasing the spatial reach. Burger et al. (2024b) investigates different scenarios for transportation, hereunder barges, vessels and pipelines within German borders, and concludes that the largest impact contributions stem from capturing and transportation. Additionally, they recommend that optimization of CCS value chains should be found in reducing the carbon intensity of the energy supply or by switching to pipeline-based transportation. We observe a similar limited number of reported impacts for injection, storage and monitoring related activities. Notably, Varling et al. (2023b) find that the modelling is quite sensitive to changes in values for injection pressures into storage sites. Based on a sensitivity analysis of one CCUS system, Leonzio et al. (2023) also suggests the storage site as important for the reduction of climate change impacts.

Examples of activities that relate to injection and storage processes could be the construction, modification or life extension of offshore platforms and interim storage facilities. Platform or site operation, here under crew shuttle, maintenance and decommissioning, could be other relevant examples considering the long operating time of CCS projects that typically range between 5 and 20 years. Offloading technologies and interfaces are not covered in any studies, although these represent a critical interface in any CCS value chain to ensure a consistent flow of CO_2 to be injected into the reservoir. Activities related to monitoring should include seismic activities as known from oil and gas exploration. Today, seismicity is most often carried out using air guns used to track plume migration of substances in the subsurface formations, which may cause harm to mammal life and generate water noise pollution. Direct emissions from seismic vessels should also be considerable for larger storage sites as previous studies have shown impacts of >20 Mt. CO_2 per year. Other technologies such as dumping seismic nodes to the seabed could be an alternative; however, only small-scale pilot demonstrations have been completed up till now. The requirement for a minimum 20 years of monitoring of offshore permanent storage sites further underpins the importance of including these processes in LCAs (Emborg et al., 2024; Gentile et al., 2024). Emborg et al. (2024) report that

preparatory and monitoring activities may account for up to 2 % of total annual emissions from operating CCS value chains at scales between 0.3 and 10 Mt. per year. The calculation approach of drilling activities may also impact the LCA results. The four studies that included drilling activities calculated impacts based on meters drilled, using the global process for offshore drilling from ecoinvent 3.9.1 and IPCC 2021 (GWP100) impact calculation method. Here, drilling a 1200-m well will result in an impact of 1193 tCO_{2eq}. A 150-day well drilling campaign (corresponding to a reservoir capacity of 2.5 Mt. CO_2) and a total climate change impact contribution of 411 tCO_{2eq} per day in drilling operation (Emborg and Olsen, 2024) result in 61,650 tCO_{2eq} (Emborg et al., 2024). Thus, the anticipated impact contribution from drilling activities could potentially be heavily underestimated. Impacts from preparatory CCS activities such as well drilling are generally deemed neglectable across the reviewed articles, outweighed by the impact contribution from capture and transport processes. However, from an improvement perspective, reducing impacts from these processes heavily relies on the realization and speed of the green fuel transition. Research and impact assessment of preparatory and storage processes, e.g. construction and operation of the interim and permanent CO_2 storage sites and CO_2 monitoring activities, may be imperative to the future optimization of CCS value chains.

In summary, to uncover the real CDR or offsetting potential of a BE/CCS project, we recommend conducting cradle-to-grave assessments while calling for the establishment of minimum boundary requirements for CCS projects that to a minimum include transport, interim storage in tanks, injection and post-injection leakage monitoring as well as drilling for well intervention, sidetracking or completion. As a comparison, the current minimum calculation requirements for CDR standard methodologies, hereunder both the EU CRCF methodology (Office of the European Union L- P, Luxembourg L, 2024) and PURO Earth (2024), distinguish between capture, transport and storage activities. Notably, these standards call for a greater level of process inclusion when assessing the life cycle impacts of CCS value chain activities than we observe in past and current literature. However, the standards still omit necessary activities implicit to most CCS projects.

5.1.2. Modelling choices

The choice of impact calculation method is occasionally found to influence LCA results potentially. Since most CCS projects are driven by the commercial opportunity for offsetting the stored CO_2, even slight differences in cradle-to-grave emissions of a CCS project may impose an economic risk to the actors involved from changes in the total CO_2 net-abatement. As of today, these standards tend to adhere to the IPCC. Thus we recommend harmonization towards the choice of impact calculation methods that are consistent with IPCC such as the EF method.

5.1.3. Functional unit

Eventually, the lack of consistency in the system boundaries and functional unit across studies limits the interpretation of the reported results as each study becomes project-specific even though CCS seeks a globally integrated deployment. Thus, we recommend a mass-based functional unit expressed with a reference flow in 'per ton of CO_2 stored' to foster harmonization and ease comparability across CCS studies. Further, we recommend that the functional unit for permanent storage of CO_2 in geological formations consistently defines (i) the captured amount within a defined operating period, and (ii) the geographical location of the project. An example of a functional unit could be "*permanent storage of 0.3 mt CO_2 per year in geological formations of the North Sea across an operating period of 15 years*". We also recommend a minimum boundary requirement that includes capture, transportation, interim storage, injection activities, and leakage and plume monitoring as well as preparatory activities related to seismic and drilling. As also highlighted by Terlouw et al. (Terlouw et al., 2021), harmonization will help unify impacts across cradle-to-grave studies of CCS value chains despite operational and infrastructural independence.

5.1.4. Temporal scope

It is often unclear whether the time horizon assumed depicts the operating period of the CCS project under study or if it frames a temporal scope intended for prospective modelling, e.g., electricity carbon intensity projection. The operating period of a CCS system may influence its global warming contribution since any extension or shortening of a project affects the accumulated as well as incremental impact contributions, e.g., depreciation of impacts related to customized infrastructure constructions across the lifespan of a project or asset. Due to the significant contribution from electricity production, the temporal location of a project may further affect the impact on climate change. From a carbon crediting perspective, the duration and temporal location of a CCS project may impact on the economic viability of CCS projects since carbon removals or offsets are estimated on an annual basis. Due to the high sensitivity related to the electricity carbon intensity and the uncertainty related to the length of the operating period of CCS projects, we call for clearer determination and delimitation of the temporal scope in LCAs of CCS projects.

5.1.5. Capture capacities and rates

Major deviations in the captured CO_2 volumes were observed. A key challenge in the conceptualization and deployment of CCS projects is the coherence between the amount of captured CO_2 and the capacity of the permanent storage site. For instance, the capture capacity influences not only the scale of capture and liquefaction plants but also transportation, e.g., vessel or pipeline sizes, as well as the size of interim storage tanks with material and energy demands increasing correspondingly. Additionally, larger, anticipated capture capacity may infringe on the storage potential of permanent storage sites even on a global scale. According to IEA, which tracks CCUS projects globally, the planned capacity by 2030 is 346.8 $MtCO_2$ per year with nearly 75 % of those capture operations taking place in Europe and North America (IEA, 2024). This may indicate that studies up till now tend to overestimate the total capacity of the value chain under study, particularly those studies considering 20 $MtCO_2$ per year or more. Unless we plan for larger interim storage sites, the capacity of the permanent storage sites may restrict capturing in the end. Since this review study did not cover point source capacity and geographical density, further studies addressing the trade-off between expanding the spatial scope of CCS projects and the environmental impacts are recommended.

No correlation between the assumed capture rate and the type of point source was observed, although this may influence the energy efficiency of capture processes. As the flue gas content has a dominant impact on the capture requirements (Brandl et al., 2021), we recommend evaluating the trade-off between the capture rate and energy requirement. Some studies compare solvents and capture schemes, typically MEA against MDEA or MEEP-based capture. For instance, Varling et al. (2023b) compare three different biogas upgrading technologies here under: MEEP-based, water scrubbers and amine scrubbers and additionally post-combustion amine scrubbers. A larger variance in the electricity demands is reported at respectively 290, 200, 28 and 110 kWh per 1000 Nm^3 biogas with supplementary heat required for the amine scrubbers with figures obtained from the literature. No direct conclusions are drawn in terms of the sensitivity related to the applied capture scheme, which seems to be generally outweighed by other technical configurations. Mio et al. (2023) reports a variable amount of CO_2 removal between MDEA absorption, membranes, and sodium hydroxide capture schemes, with membranes achieving 95 % removal at best. Notably, no study reports on solvent losses nor impacts related to the degradation of solvents, except from Mio et al. (2023) and Huo et al. (2022), which report respectively on MEA solvent degradation and regeneration. As the technological readiness level of different capture schemes improves, we recommend further research with a focus on comparing capture schemes and quantifying the potential difference in environmental impacts, including sensitivity related to electricity consumption, trade-offs from additional heat requirement, and the impacts

of solvent degradation and regeneration not covered in today's literature.

5.1.6. LCA impact category coverage

A larger deviation in impact category coverage across the reviewed studies is observed spanning from studies only reporting on the climate change category to all impact categories as recommended by the European Commission (Directorate-General for Environment, 2021). In addition, the studies reported differences in relation to increases or decreases within each category. For instance, Facchino et al. (2022) concluded a need for a holistic-oriented approach in the design of CCUS configurations to avoid burden shifts throughout the value chain. In all scenarios, climate change was decreased, whereas abiotic depletion, eutrophication, acidification and ozone depletion increased, attributed to a higher requirement of natural gas to compensate for the electricity demand of the capture unit. Hospital-Benito et al. (2023) also highlighted the importance of evaluating the environmental burden of ionic liquid-based (IL) CCS processes compared to conventional amine-based systems. Results showed that ILs are more harmful to the environment than amines. Yet, the life cycle impact assessment carried out for both CCS processes showed lower environmental impact values for all estimated impact categories for IL compared to MDEA due to an almost 10 % energy-efficiency performance. These observations underline the importance of applying a holistic-orientated approach towards LCA of CCS systems and technologies with the inclusion of other impact categories apart from climate change. Similarly, Müller et al. (2020) and Boakes et al. (2023) underline the importance of including other impact categories than climate change to avoid misinformed decision-making. The Wolf et al. (2010) has outlined a set of recommendations on the use of Environmental Footprint methods (Directorate-General for Environment, 2021). Based on this, we recommend the EF method as impact calculation method and a minimum inclusion of the sixteen impact categories referenced for those LCAs that aim to assess the full environmental impact of CCS systems. In those cases where LCA is applied to assess the climate change impacts i.e., carbon footprint study, of CCS systems intended towards carbon removals or offsets, we recommend the EF method and the climate change impact category at a 100-years horizon to align with CDR standard methodologies such as PURO Earth (2024) and the EU's CRCF draft methodology (Office of the European Union L- P, Luxembourg L, 2024).

5.1.7. Electricity projections

The potential changes in the European energy system challenge comparability and interpretation of the climate change benefits linked to green fuel transition across studies. Furthermore, from the studies applying energy scenario modelling, it was not always evident to what extent the projections followed the anticipated operating period of the CCS project under study. Thus, the timing of when climate change reductions are realized is unclear. Similar lack of transparency with respect to modelling choices and background system characterization has been previously flagged when LCA is combined with future scenario analysis (Bisinella and Christensen, 2021; Arvidsson and Molander, 2017). Since most BE/CCS projects (globally) are in the initial or early stage of planning or construction (IEA, 2020), and due to the long prosperity of CCS projects, typically not planned for <35 years of operation, specifying characteristics of the functional unit to allow for variation over time is crucial, as technology and solutions may change during these longer time horizons. Considering the large sensitivity related to the carbon intensity of electricity production, we recommend that energy scenario modelling relies on an incremental projection per year throughout the operating period, and not on a high-low or average scenario basis. This is prudent in CDR or carbon offsetting estimations yet also in terms of accurately estimating the amount of carbon to be stored, i.e., realization of said positive contributions to climate change reductions. To lower uncertainty and promote consistency in the reported LCA results, we recommend harmonization of projection factors for electricity emission intensities and energy scenario modelling by defining a set of reference factors to be generally applied.

5.1.8. Comparison against theoretical values

Regarding the reported energy demands, a major consequence of the typical granularity level in current life cycle inventories of CCS is that only a few studies report values that can be compared against theoretical estimates. This underpins the need for higher process input specificity to foster comparability and strengthen the basis for uncertainty evaluation in selected designs of CCS value chains. Since direct comparison between the LCAs is already challenged by the differences in general study and LCA characteristics, a comparison of input values for energy requirements of selected CCS processes against theoretical values can be prudent and useful way for evaluating the effectiveness of a CCS project. This will help to reflect the uncertainty associated with a project's reported CO_2 removal or offsetting potential. Eventually, not all CCS processes were covered in this review; hence, we recommend that similar theoretical calculations be established, e.g., vessel transport and monitoring (depending on the chosen technology). Additionally, further research is encouraged, evaluating currently planned CCS projects to assess the feasibility of the claimed efficiencies.

5.2. Recommendations

In summary, large differences were observed in the LCA outputs although comparability across studies was challenged by the lack of harmonization in scope, boundaries, functional units and modelling choices, which are also in line with the conclusions made by Cuéllar-Franca and Azapagic (2015), Müller et al. (2020), and Zimmermann et al. (2022). LCA is a standardized method under ISO 14040/44 (International Organization for Standardization, 2006a; International Organization for Standardization, 2006b) which helps ensure consistency in the application of the LCA methodology framework. However, the standard leaves room for methodological choices such as functional unit definition, system boundary setting, background and foreground processes and impact calculation methods. In line with the conclusions of Lerche Raadal and Saur (2022), the choice of modelling decision may challenge the comparison of the generated results while underpinning the importance of consistent system boundaries. We call for a standardized minimum boundary requirement to increase comparability across LCA studies and to strengthen the validation of reported results. In the context of LCA of CCS projects in the planning phase, we recommend higher granularity through equipment-specific energy demands. This to allow for validation of any anticipated energy demands through comparison against thermodynamic values as demonstrated in Section 3.4 based on the calculations provided in SM2, and our observations shown in Section 4.4.

Below presents our recommendations regarding the application of LCA as methodology for quantifying the greenhouse gas emission of CCS systems, particularly with the purpose of strengthening comparability and standard methodologies for carbon offsetting:

1. A mass-based functional unit for permanent storage of CO_2 in geological formations expressed with a reference flow in 'per ton of CO_2 stored' which defines (i) the captured amount within an operating period, and (ii) the geographical location of the project.
2. Minimum boundary requirements besides capturing including transportation, interim storage in tanks, injection, leakage and plume monitoring, and preparatory activities related to seismic and drilling.
3. Coverage of impact categories in line with the sixteen impacts categories recommended by the European Commission (Directorate-General for Environment, 2021)
4. Harmonization of impact calculation methods that are consistent with IPCC e.g., the EF method.

5. Total and incremental reporting of CO_2 net-storage volumes to determine timely realization of removals and offsets.
6. Harmonization of projection factors for electricity emission intensities and energy scenario modelling by defining a set of reference factors to be generally applied.
7. Reporting of energy demands for capture processes with distinguishment between electricity use, heat generation, compression and liquefaction.
8. Providing the following design parameters and technical specifications for:
   - Capturing (capture rate, reboiler heat duty and any heat loss),
   - Compression (compression stages, mechanical and electrical driver efficiency),
   - Liquefaction (COP value and CO_2 pressure and temperature),
   - Pipeline (mass of carbon steel and chromium content (if any), pressure drops per unit length, maximum pressure, and the number of pressure boosters),
   - Truck transport (gross weight and average load),
   - Injection (injection pressure rate, energy type),
   - Drilling (days or meters drilled, mass of casing and chromium content (if any),
   - Leakage monitoring (distance sailed by seismic or supply vessel)

5.3. Study limitations and future research

The modelling choices (i.e., process-based, input-output or hybrid) and quality of the LCA (i.e., compliance with the ISO standard (International Organization for Standardization, 2006a; International Organization for Standardization, 2006b)) of the presented literature were not evaluated in this study. Whilst result uncertainties may be linked to the modelling and quality of the LCA studies (Bisinella and Christensen, 2021; Mendoza Beltran et al., 2018; Niero et al., 2015), further studies thoroughly evaluating the assignment of uncertainty to the input data are recommended.

In the early stages of the study selection, it became apparent that a vast majority of articles centered on CCU or CCUS with generally less specificity on CCS. The geographical location of the reviewed CCS projects was not evaluated as part of this review, yet it was shown to significantly impact energy scenario modelling and the total impacts of a CCS project, mainly related to the energy source locally available or distances for CO_2 transportation. Based on the observed difference in the reported carbon intensities for electricity across the studies and the high sensitivity for the capturing and liquefaction process, further work is recommended to focus on the energy projection of CCS projects.

The carbon penalty of CCS has been addressed in several environmental studies throughout the past two decades. Despite an increase in scientific publications, the proportion of papers addressing CO_2 transportation and the related infrastructure, interim and permanent CO_2 storage operations as well as CO_2 monitoring is still low. Overall, there is a dominance of carbon capturing technology-related research, despite the increase of LCA application to assess full value chain life cycle impacts for CCS projects. Few studies exist on carbon infrastructure, injection operation and storage activities, and often those studies that do focus on carbon transportation and storage, do not provide a detailed process-level insight. During the early development of CCS value chains, several factors have been shown to cause potential impacts and limitations to the selection of technologies and conceptualization of CCS projects. Existing infrastructure, available means of transportation, storage site location, reservoir conditions and capacities are among the factors that may limit designers in choosing the optimum solution. Surprisingly, and despite the underlined importance of scaling up CCUS value chains both from a capturing and storage capacity level (IEA, 2020), none of the reviewed studies addressed scale-up scenarios. Moreover, as fossil-based energy production is phased out, the excessive impact burden will shift from energy-intensive processes such as amine capture and compression. Efficiency in CCS value chains will thus be found in those parts of the chain less covered in literature today, e.g., monitoring, injection operations, well drilling, infrastructural efficiency, and other downstream aspects of the value chain.

Land use was not assessed as part of this review, but a relatively large land occupation linked to the capture plant and interim storage facilities has previously been reported. For instance, Emborg et al. (2024) report an accumulated area footprint for onshore storage ranging from 1500 to 15,000 m^2 for a single facility depending on the annual capacity between 0.3 and 10 Mt. CO_2 per year. Thus, we encourage further research on the impact related to land occupation for capture, liquefaction and interim storage facilities. Moreover, few studies report on the impacts related to solvent loses or degradation of solvents, hereunder, Gentile et al. (2024), Facchino et al. (2022), Bello et al. (2020) and Yang et al. (2019). As the technological readiness level of different capture schemes improves further research required focusing on the comparison of capture schemes including quantifying potential impact differences and sensitivity.

6. Conclusion

This review analysed the life cycle assessment (LCA) of 33 scientific publications assessing various options for carbon capture and storage (CCS). A systematic literature review approach was followed with the objective of gaining a better understanding of how LCA is used for the assessment of the environmental impacts of CCS systems. The alleged effects of CCS systems were critically assessed through a comparison of reported life cycle inventories and the result of the life cycle impact assessment as well as a comparison of reported energy demands against theoretically estimated values within CO_2 capture, transportation and storage processes.

Most of the reviewed studies focus on capture from post-combustion during industrial processes from waste-to-energy, cement or steel production, or from natural gas or coal-fired power plants. Large ranges in anticipated capture capacities are observed with a tendency to overestimate storage capacities; moreover, there seem to be a risk of overestimating capture capacities compared to the reported estimated storage capacities. In fact, no study reported the storage capacity of the storage site envisaged for the case study. A tendency to assume a 90 % capture rate is observed without justification; moreover, no correlation between the assumed capture rate and the type of point source was reported although this may influence the energy efficiency of capture processes significantly. Less than half of the studies defined a temporal scope. Generally, the completeness and reporting of life cycle inventories is poor. Processes related to interim storage, injection, permanent storage, monitoring and preparatory activities such as drilling are generally neglected or excluded. Operating conditions and specifications for vessels and pipelines are generally missing despite the transportation of CO_2 via pipelines are frequently suggested for optimization, as well as the transition towards alternative fuels for vessels. Preparatory activities, namely drilling of injection wells, were limited with larger uncertainties observed in the reported results. Typically, process inputs are covered by energy requirements, lacking material specification for facilities and constructions as well as waste generated, hereunder, MEA degeneration. The functional unit is often defined differently yet typically based on either a mass-based or energy-based unit. The most frequently considered midpoint impact category is climate change followed by acidification and freshwater eutrophication. Almost all the reviewed studies report a positive contribution to reducing climate change yet at the expense of additional impact categories.

Several methodological choices in LCA studies may significantly influence results; the choice of impact calculation method differs with a slight preference to apply ReCiPe 2016, EF 3.0 and CML 2001, further reducing the comparability of the results. Despite large water consumption in MEA capture schemes, impacts related to water use are rarely covered. The amount of net-stored CO_2 range between 27 and 98

%. Little consistency in system boundaries and functional units across studies limits the interpretation of the reported results. Consequently, each LCA study becomes incomparable even though CCS seeks towards a globally integrated deployment.

The review summarizes a set of recommendations centered on evaluating the trade-off between the capture rate and energy requirement from an environmental perspective. With high sensitivity related to electricity consumption and the electricity carbon intensity factor in combination with uncertainty related to the estimated operating period of CCS projects, greater emphasis on the temporal scope of CCS projects is recommended. As the technological readiness level of different capture schemes improves, further research will benefit from a focus on the comparison of capture schemes and the differences in impact contributions. Hereunder, the sensitivity related to electricity consumption, trade-offs from additional heat requirement, and the impacts of MEA solvent degradation which are not yet well covered in the literature. Comparison of the reported input values for energy demand for CCS processes against theoretical estimated values showed deviations within both capturing, compression, liquefaction, pipeline transport and injection processes with an overweight of underestimated energy demands, which may mean that current CCS projects may be undersized or unrealistic in their assumptions regarding energy requirements. However, this comparison was generally challenged by low granularity in the life cycle inventories while higher specificity of energy inputs for each process is urged to obtain greater comparability and transparency in the reported results.

In response to the challenges highlighted, and with the objective of strengthening future LCAs of CCS projects towards quantification of carbon offsets or removals, it is recommended to include:

- A cradle-to-grave perspective.
- A common, mass-based functional unit with the reporting reference flow ‘per ton of CO_2 stored’.
- Establishment of minimum boundary requirements with the inclusion of the following unit processes: capturing, compression, transport, interim storage, injection and monitoring as well as drilling, with optional boundaries to include liquefaction and other preparatory activities such as platform modification and seismicity.
- A minimum level for input flows to cover energy and material demands for each unit process.
- A minimum level for the reporting of design parameters and technical specifications, to strengthen comparability and validation of reported impacts and claimed storage volumes.
- Sensitivity analysis for the energy demand for capture processes.
- Projection of the electricity carbon intensity based on annual projections covering the full operating period of the CCS project, and not on a high-low or average scenario basis, to understand when removals or offsets are realized.
- Validation of input values for energy demands, for CCS projects in the planning stage to assess the uncertainty related to anticipated energy requirements, through application of the demonstrated mechanistic modelling approach.

In conclusion, although the deployment of CCS systems seeks to mitigate climate change and the fact that most studies report overall positive contributions to climate change, some uncertainty relates to when and to what extent reductions are realized. A lack of standardized guidelines for system boundaries and the operating period as well as the timing of deployment of the different CCS systems, is evidently shown through the mis-harmonization in depicted time horizons, capture capacities and processes included across the reviewed studies. Eventually, these uncertainties may also challenge the evaluation of carbon removals and offsets, which are integral to commercialization of most CCS projects.

CRediT authorship contribution statement

M. Emborg: Writing – original draft, Visualization, Validation, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **L. Kisieliene:** Writing – review & editing, Methodology, Formal analysis, Data curation. **N. Thonemann:** Writing – review & editing, Methodology. **A.A. Eftekhari:** Writing – review & editing, Supervision, Software, Methodology, Data curation. **S.I. Olsen:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Mia Emborg reports financial support was provided by Innovation Fund Denmark. Mia Emborg reports financial support was provided by INEOS Energy DK. Mia Emborg reports a relationship with INEOS Energy DK that includes: employment and funding grants. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.spc.2025.09.015.

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